

# BFS Traversal

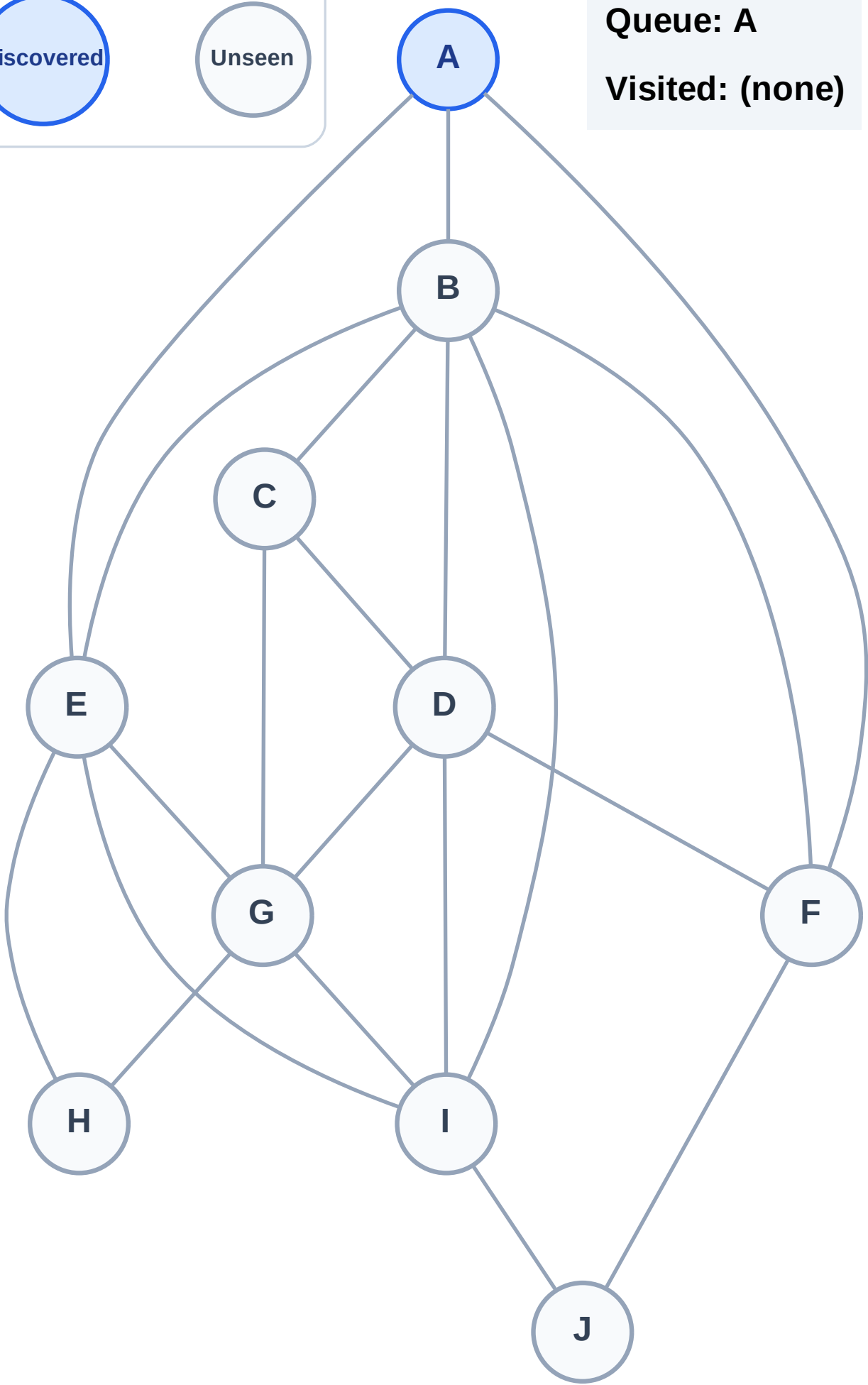
Step 1: Start at A

## Legend



Queue: A

Visited: (none)



# BFS Traversal

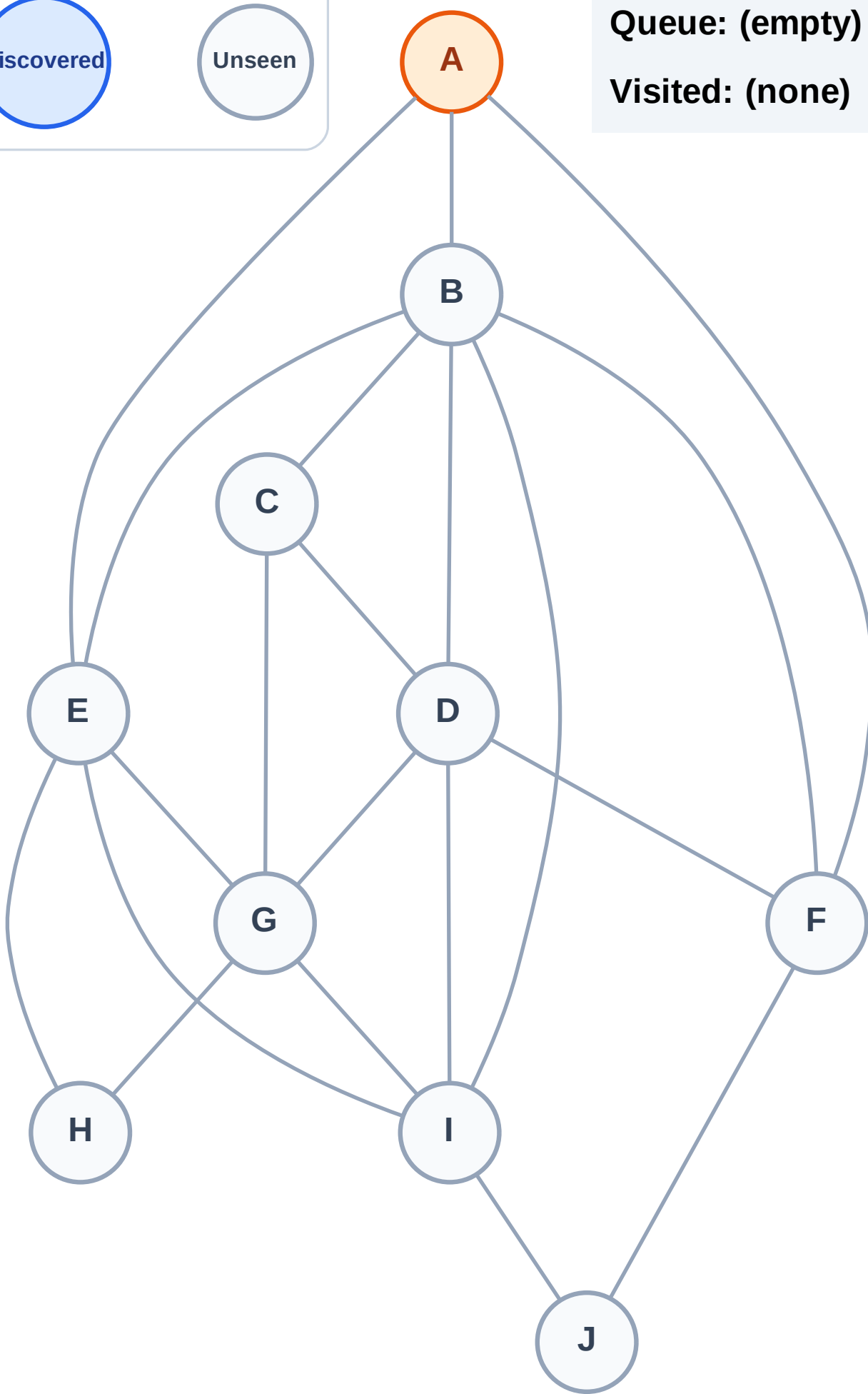
Step 2: Process node A

Legend



Queue: (empty)

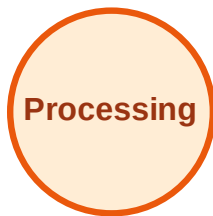
Visited: (none)



# BFS Traversal

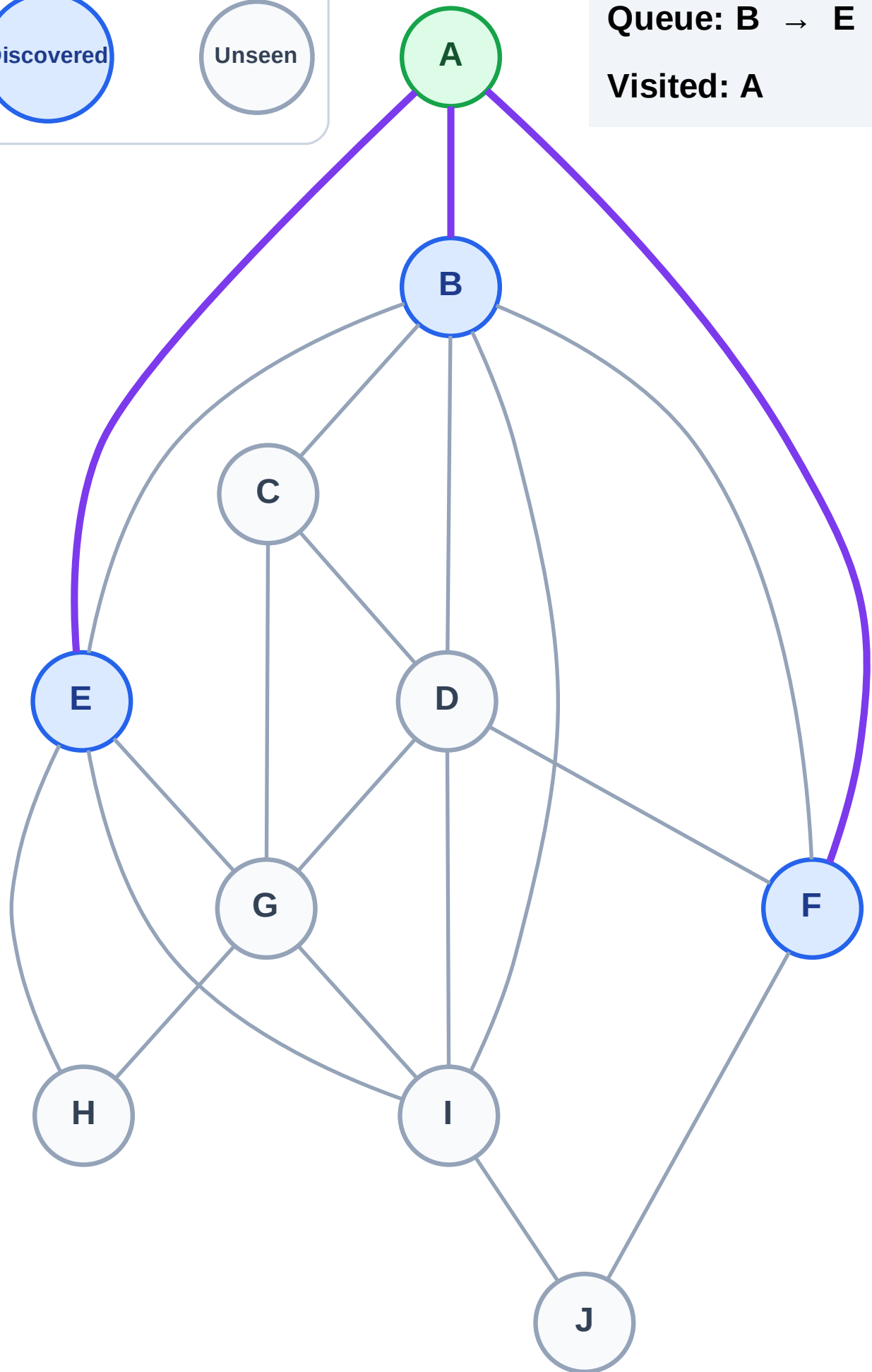
Step 3: Discover B, E, F from A

## Legend



Queue: B → E → F

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

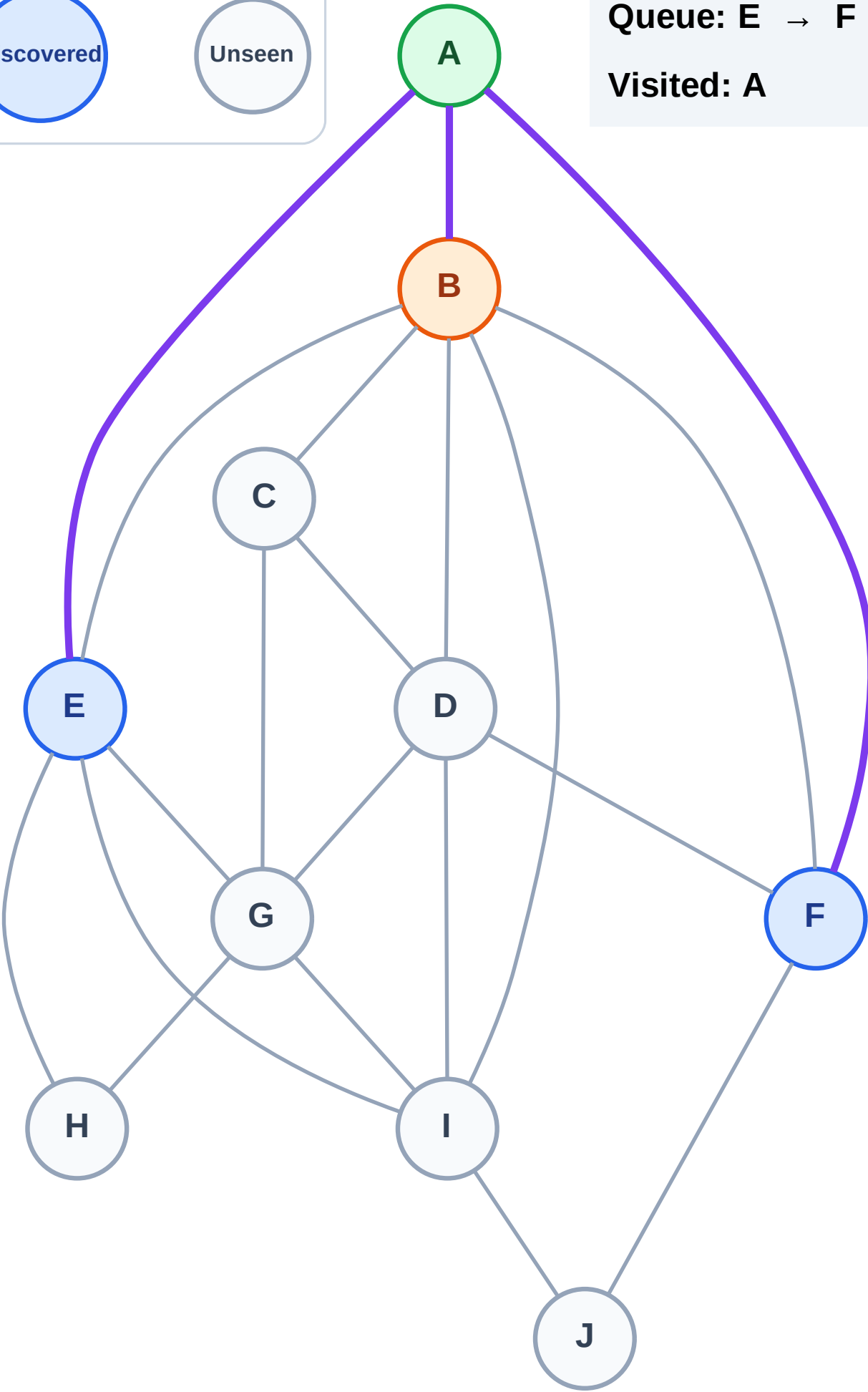
Processing

Discovered

Unseen

Queue: E → F

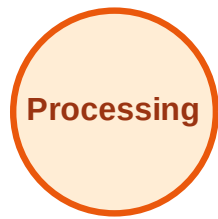
Visited: A



# BFS Traversal

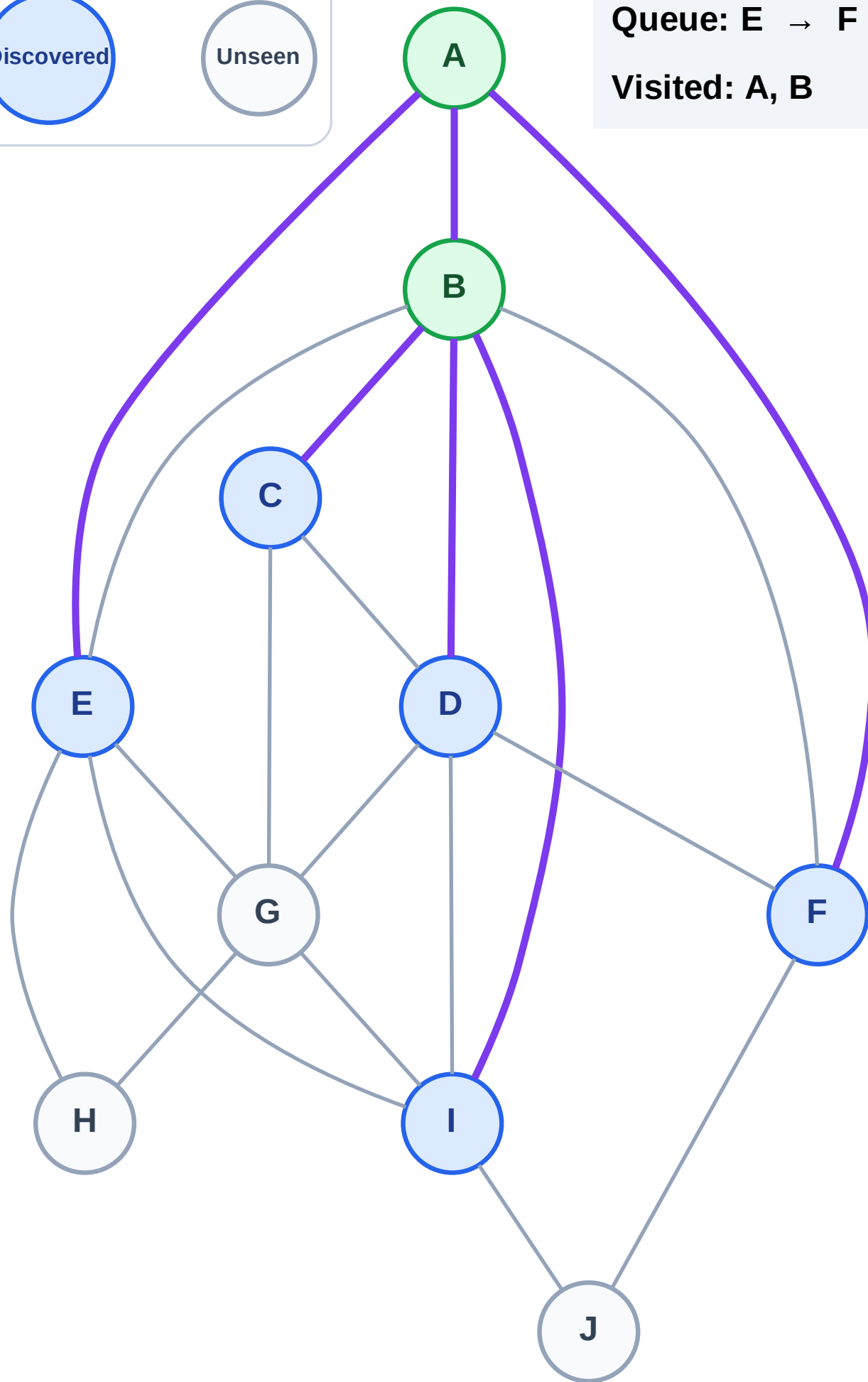
Step 5: Discover C, D, I from B

## Legend



Queue: E → F → C → D → I

Visited: A, B



# BFS Traversal

Step 6: Process node E

Legend

Complete

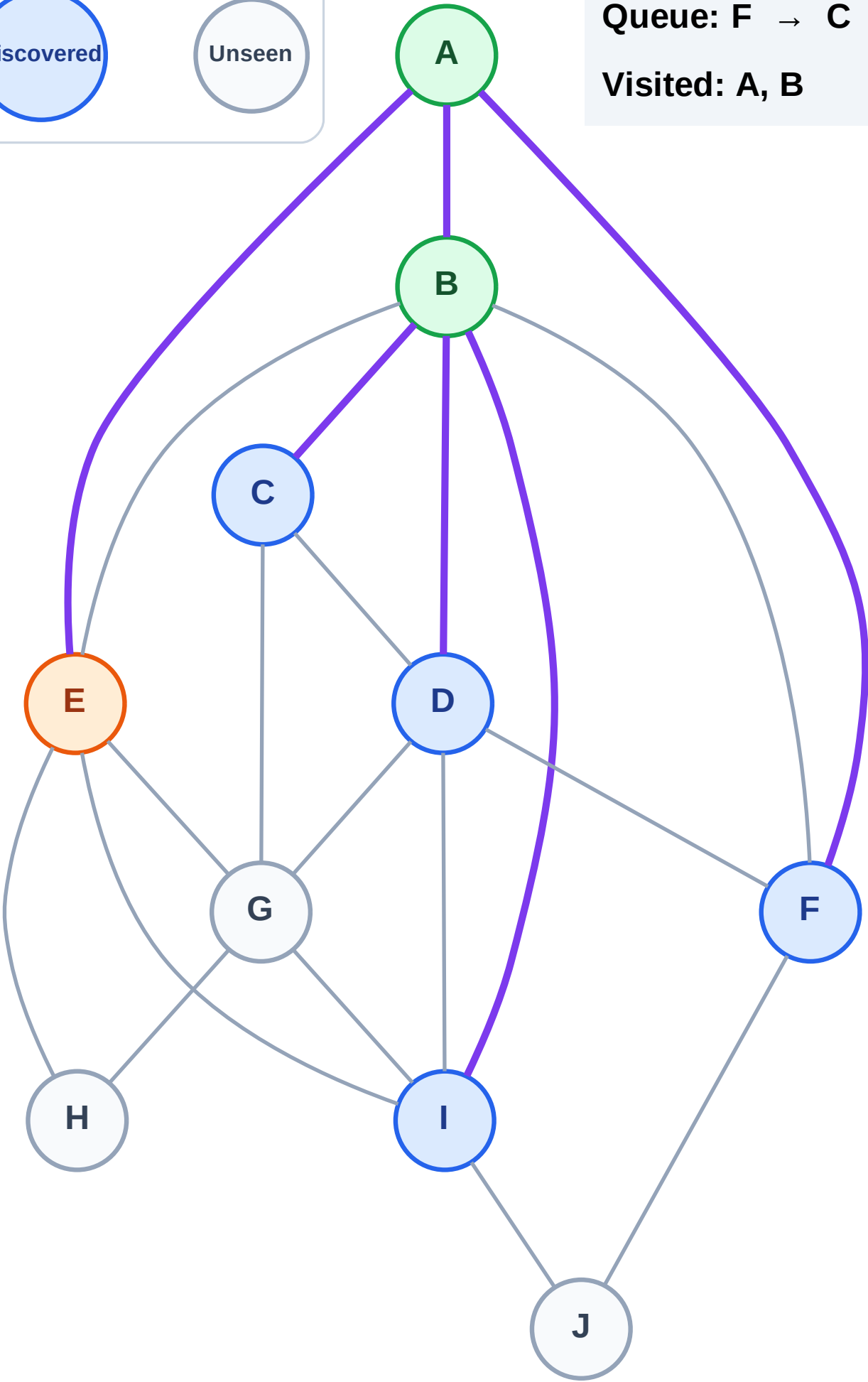
Processing

Discovered

Unseen

Queue: F → C → D → I

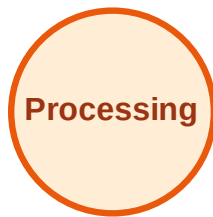
Visited: A, B



# BFS Traversal

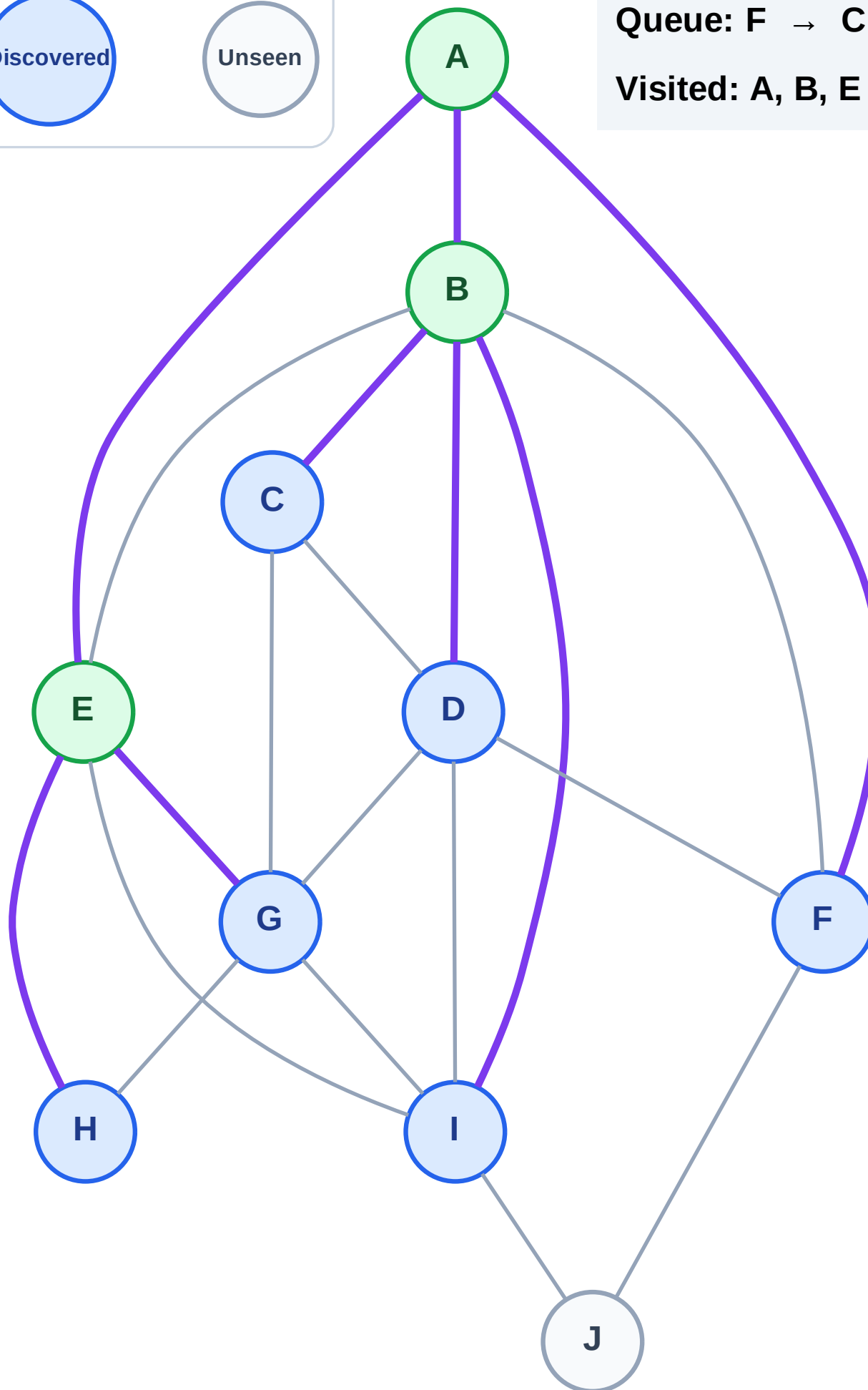
Step 7: Discover G, H from E

## Legend



Queue: F → C → D → I → G → H

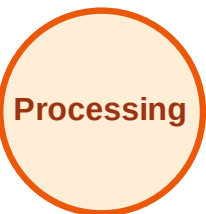
Visited: A, B, E



# BFS Traversal

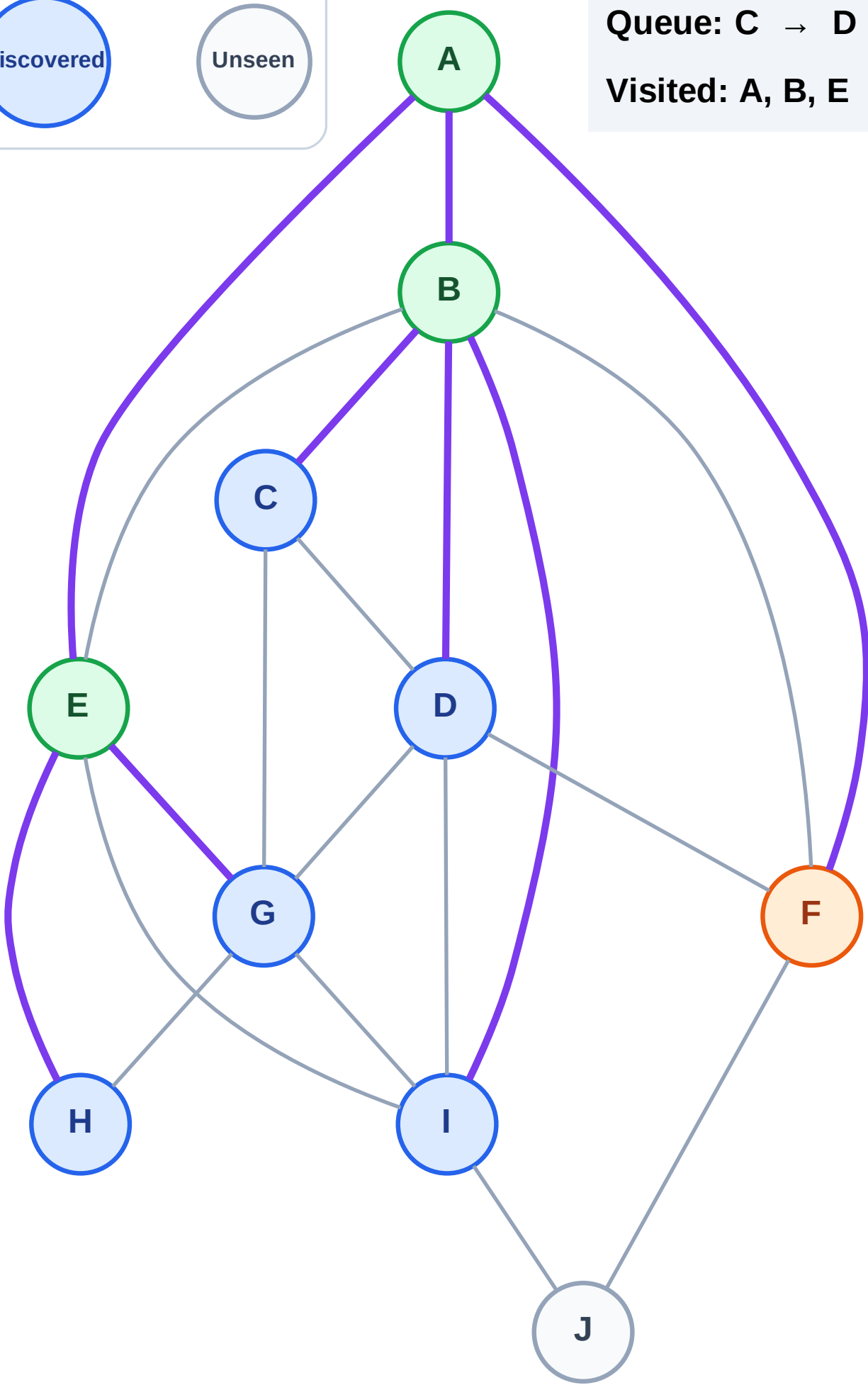
Step 8: Process node F

## Legend



Queue: C → D → I → G → H

Visited: A, B, E



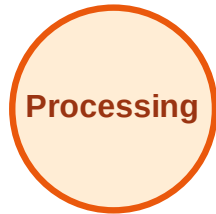
# BFS Traversal

Step 9: Discover J from F

## Legend



Complete



Processing



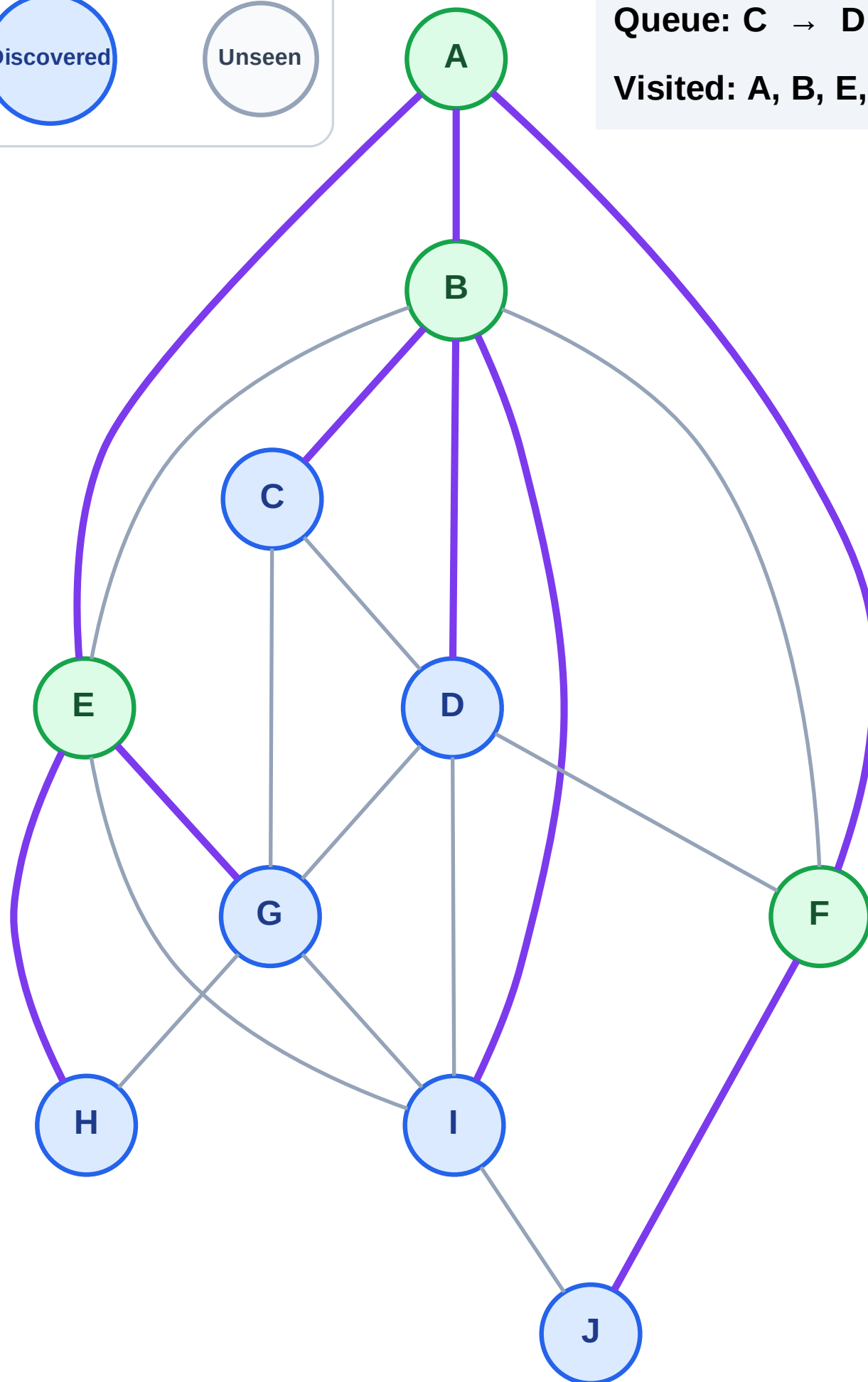
Discovered



Unseen

Queue: C → D → I → G → H → J

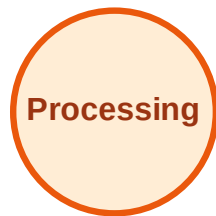
Visited: A, B, E, F



# BFS Traversal

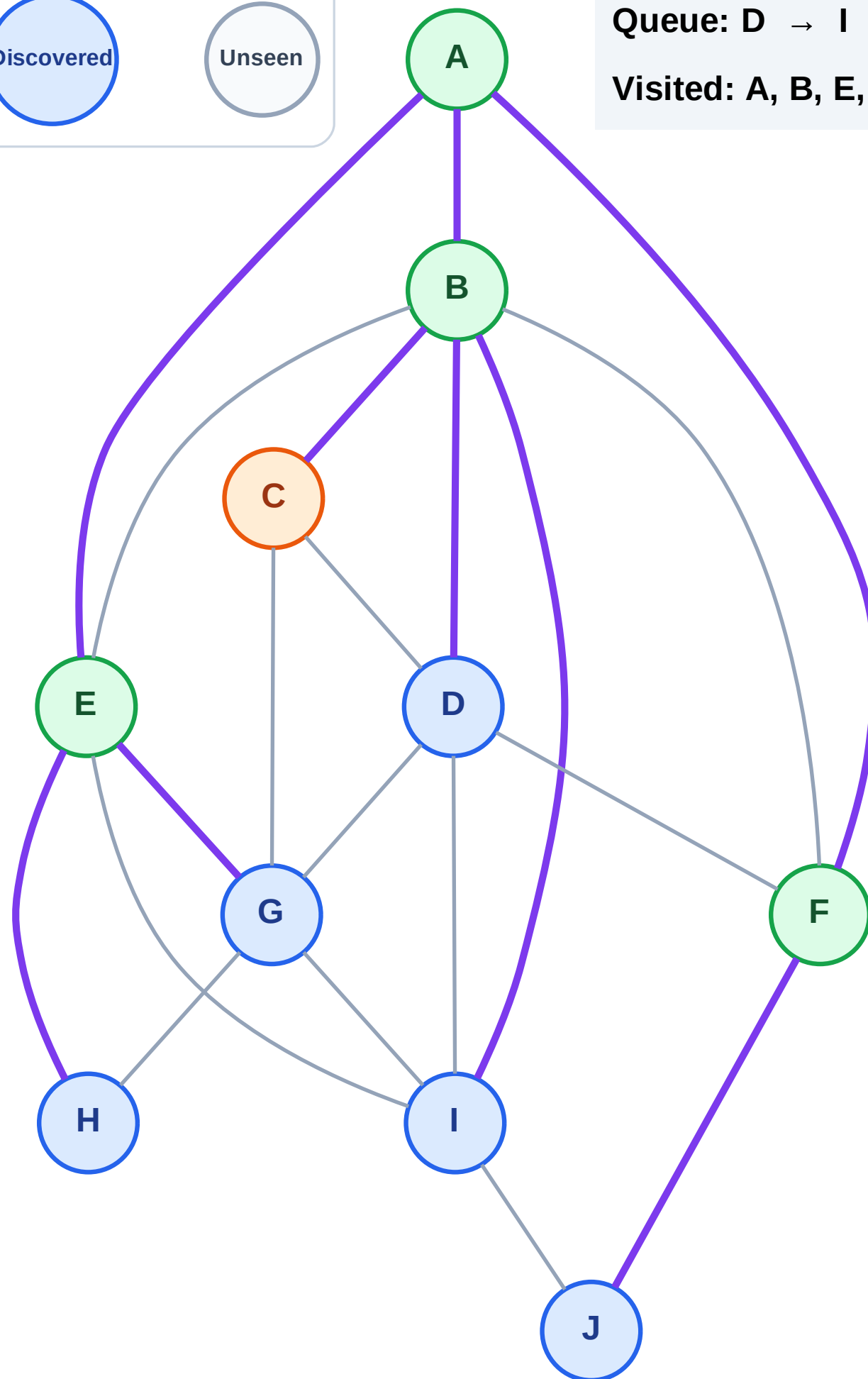
Step 10: Process node C

## Legend



Queue: D → I → G → H → J

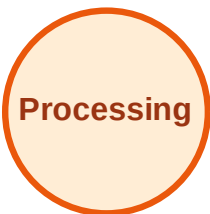
Visited: A, B, E, F



# BFS Traversal

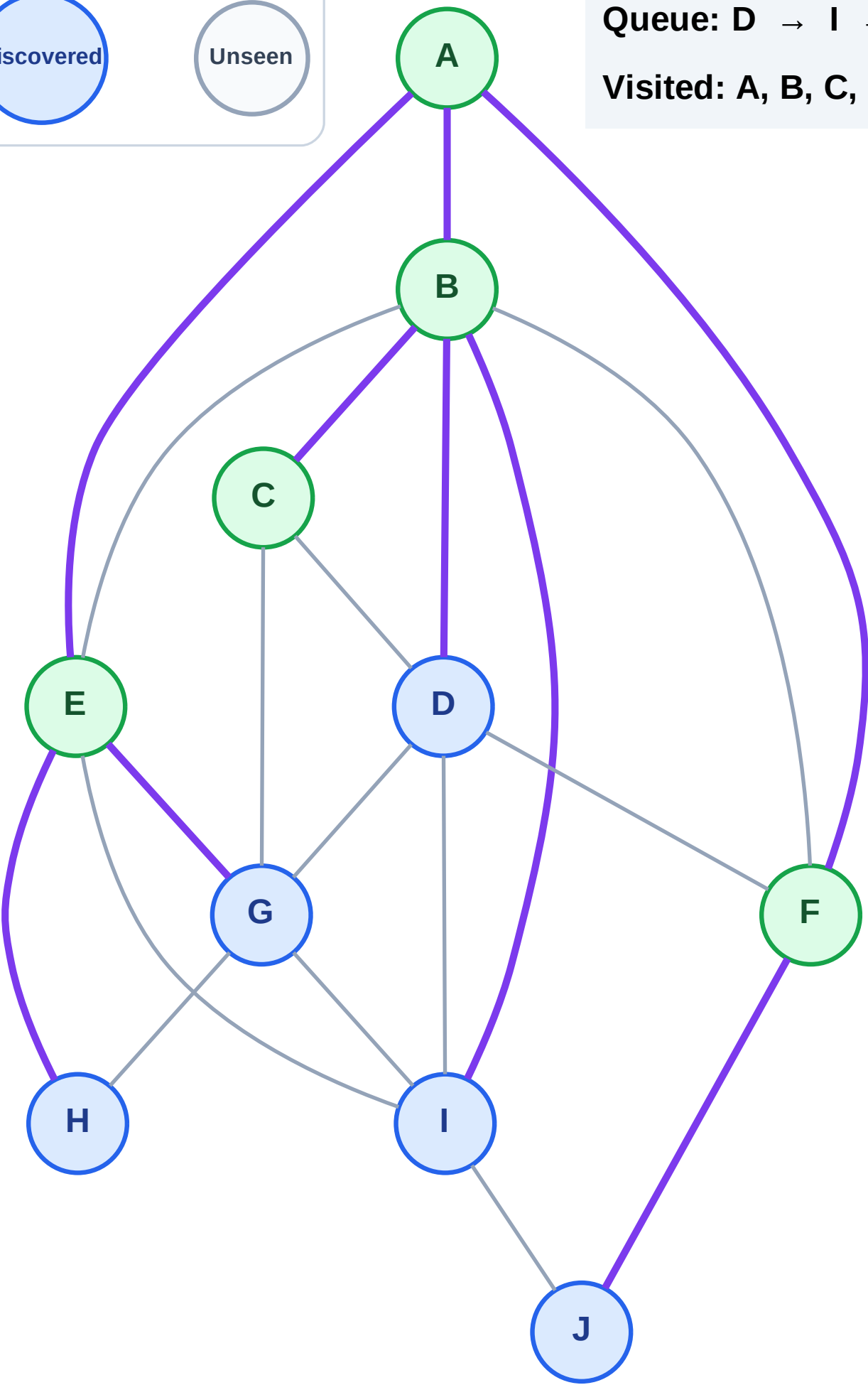
Step 11: No new neighbors from C

## Legend



Queue: D → I → G → H → J

Visited: A, B, C, E, F



# BFS Traversal

Step 12: Process node D

Legend

Complete

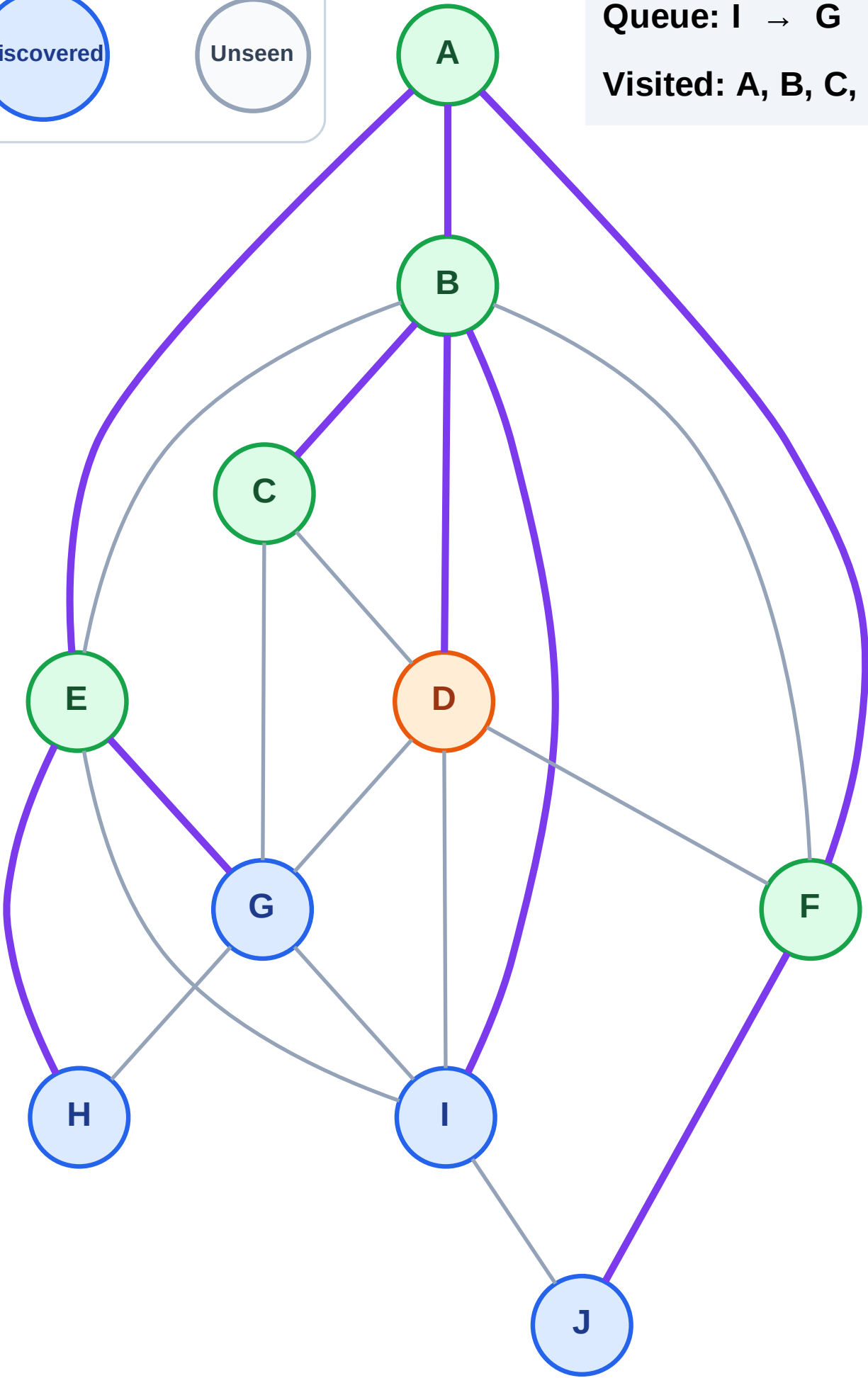
Processing

Discovered

Unseen

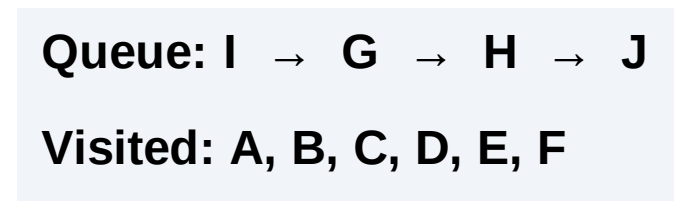
Queue: I → G → H → J

Visited: A, B, C, E, F



Step 13: No new neighbors from D

Step 13: No new neighbors from D



# BFS Traversal

Step 14: Process node I

Legend

Complete

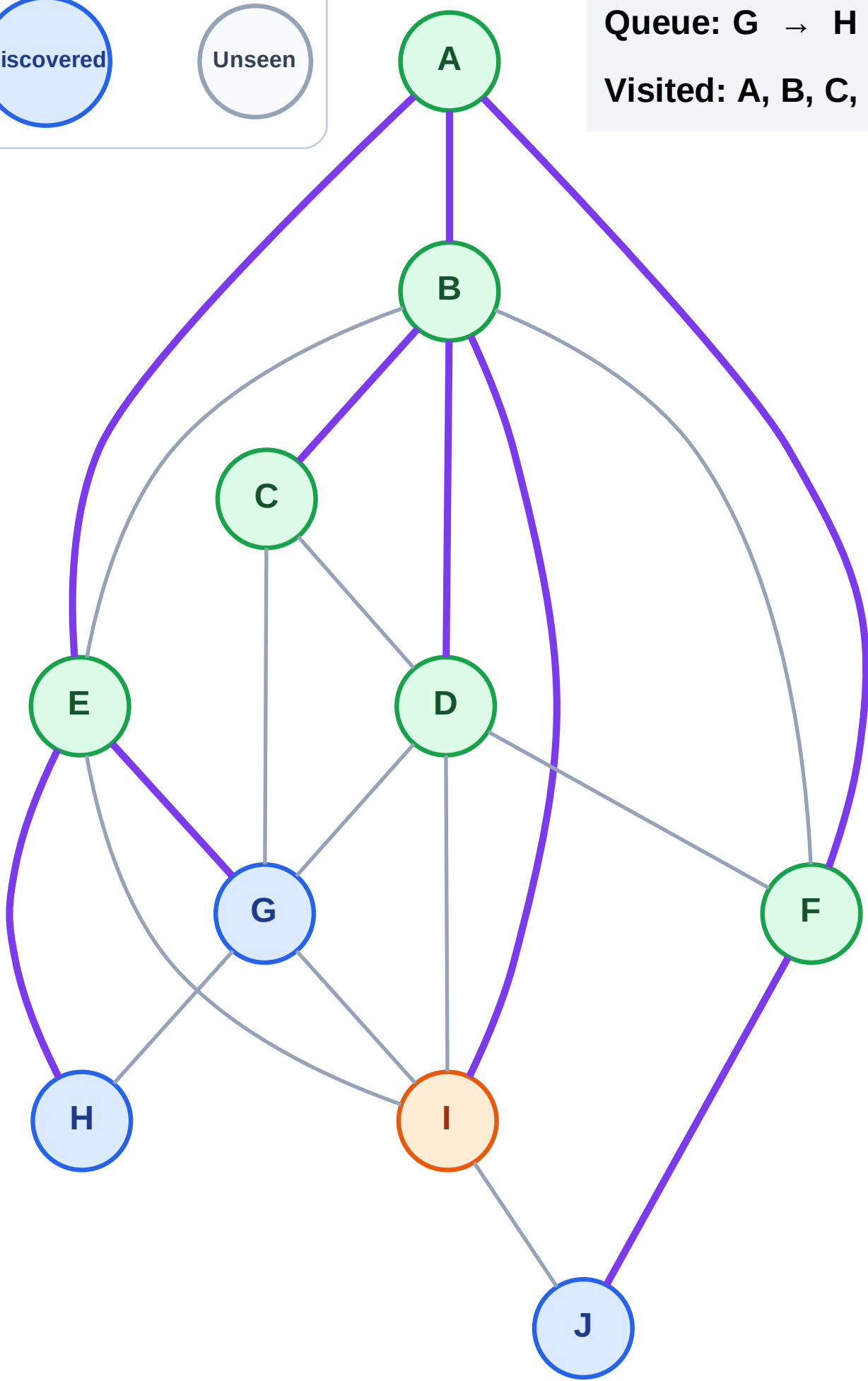
Processing

Discovered

Unseen

Queue: G → H → J

Visited: A, B, C, D, E, F

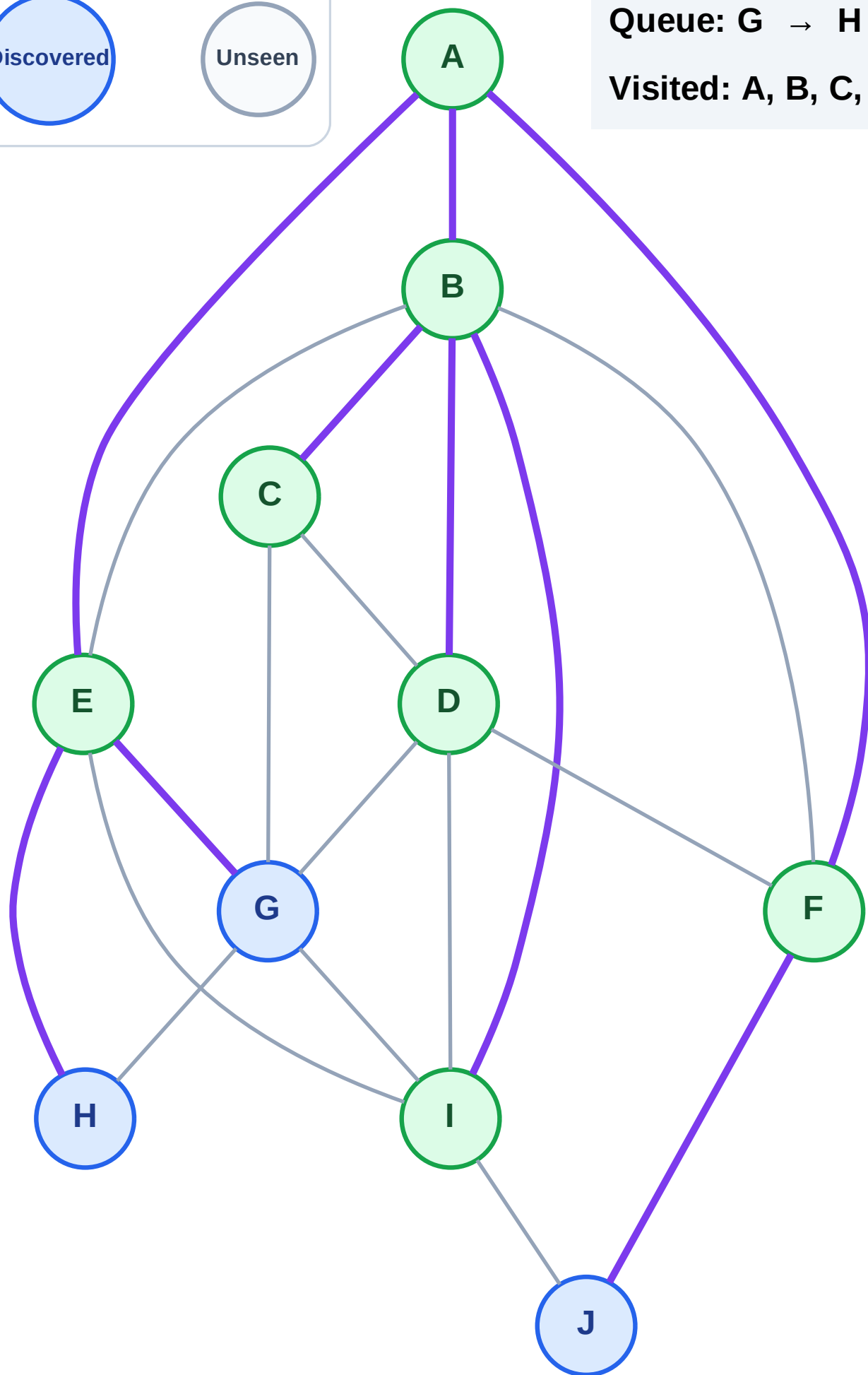


Step 15: No new neighbors from I

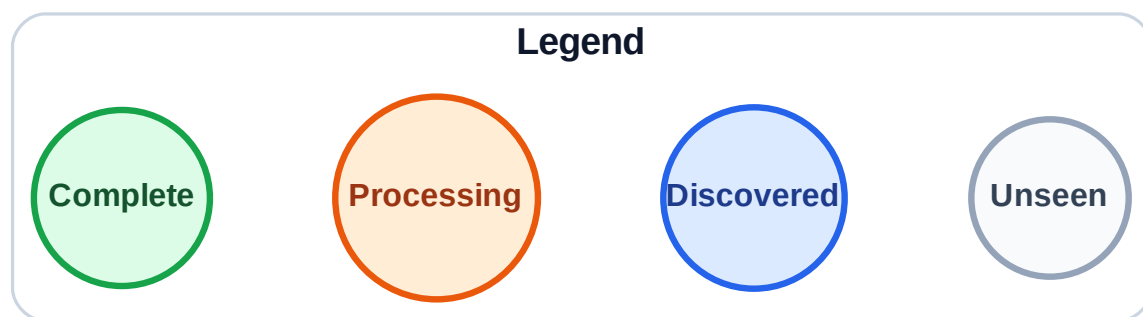
## Legend

## Unseen

**Visited: A, B, C, D, E, F, I**

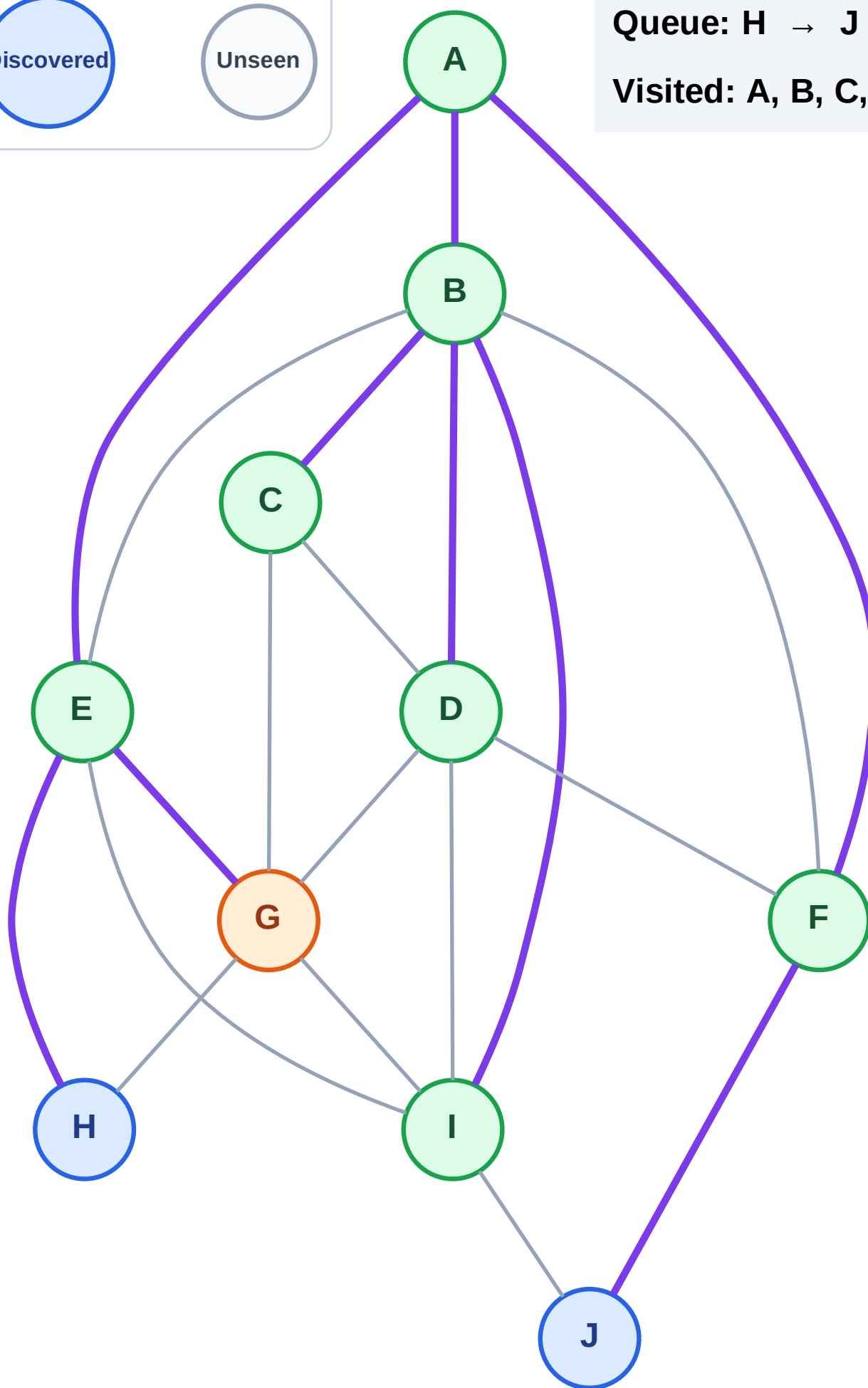


## Step 16: Process node G



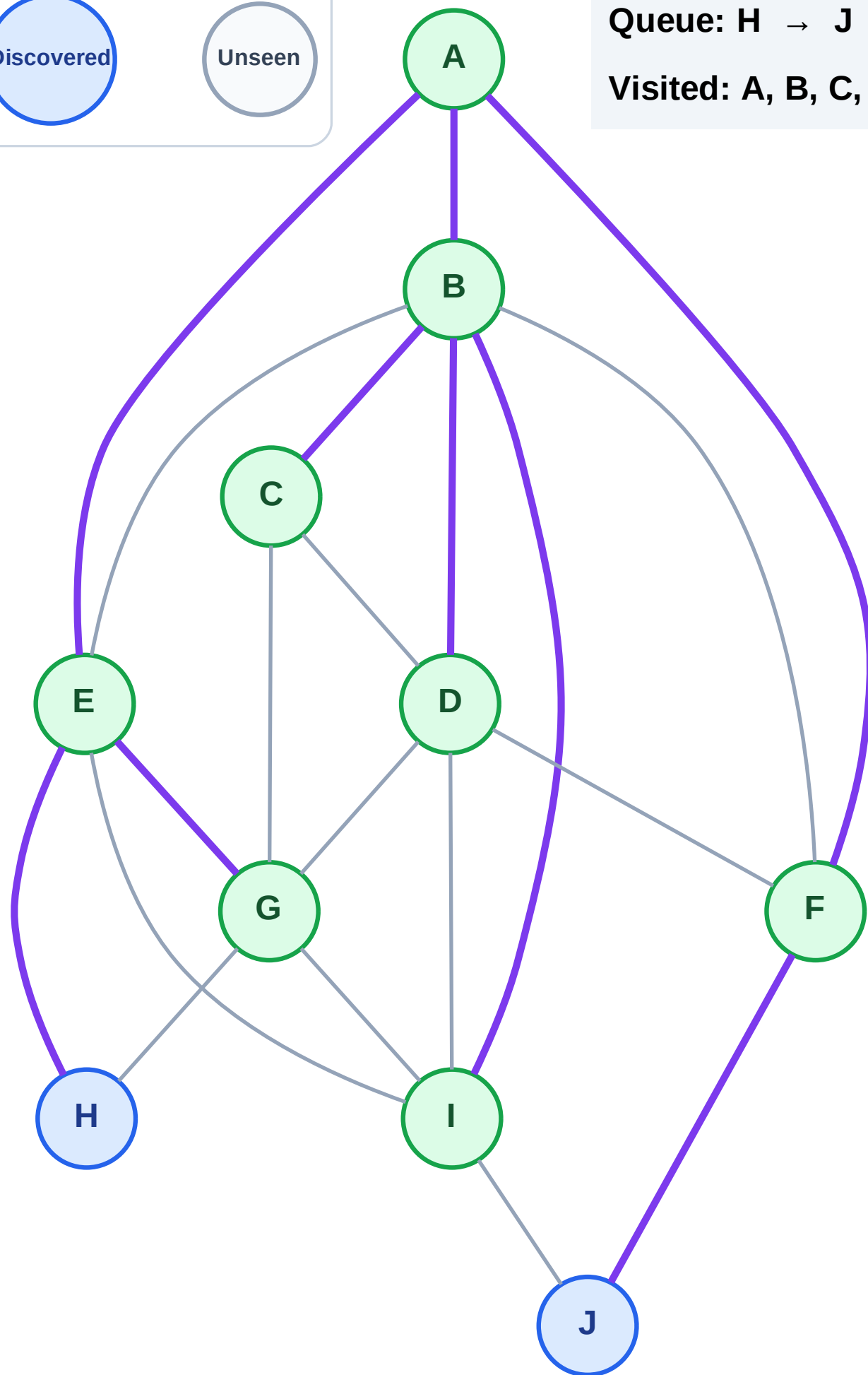
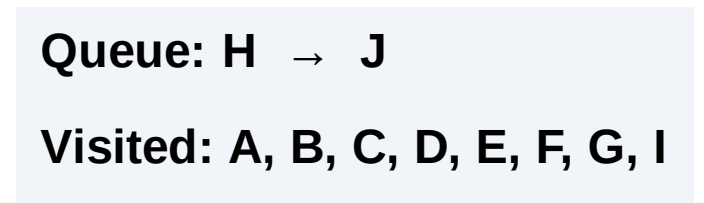
**Queue: H → J**

**Visited: A, B, C, D, E, F, I**



Step 17: No new neighbors from G

Step 17: No new neighbors from G



# BFS Traversal

Step 18: Process node H

## Legend

Complete

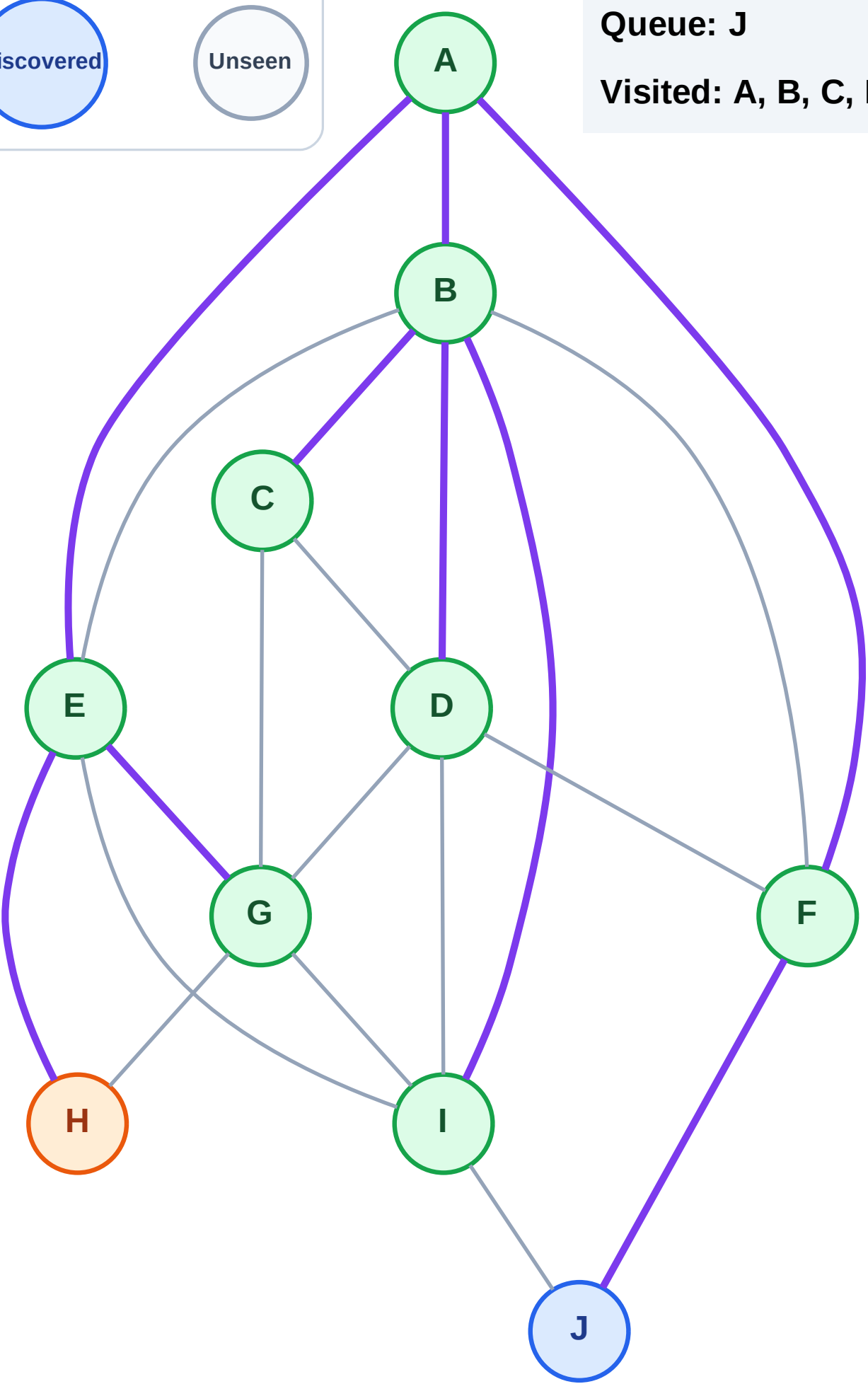
Processing

Discovered

Unseen

Queue: J

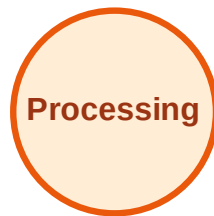
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

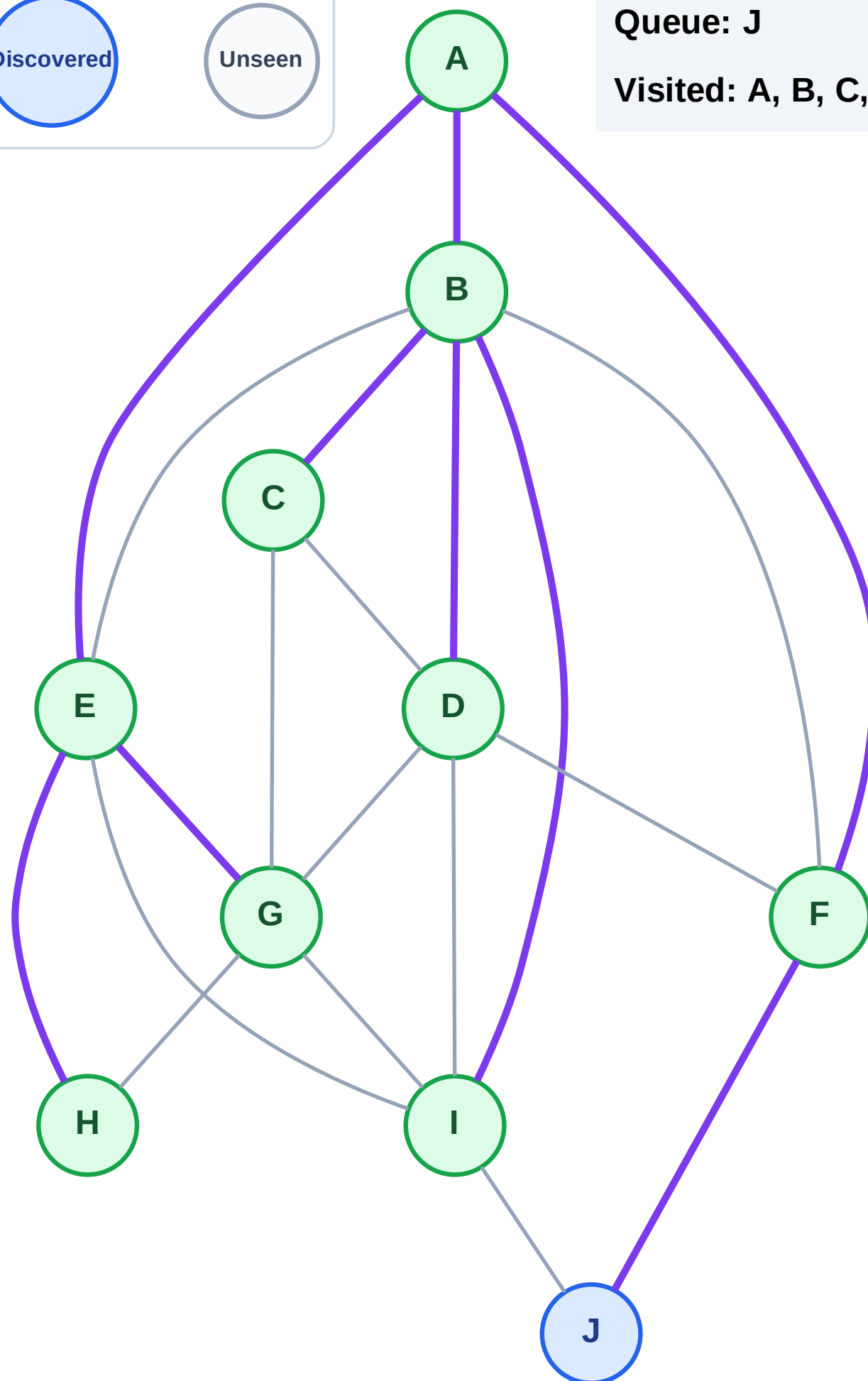
Step 19: No new neighbors from H

## Legend



Queue: J

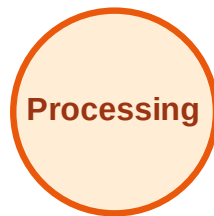
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

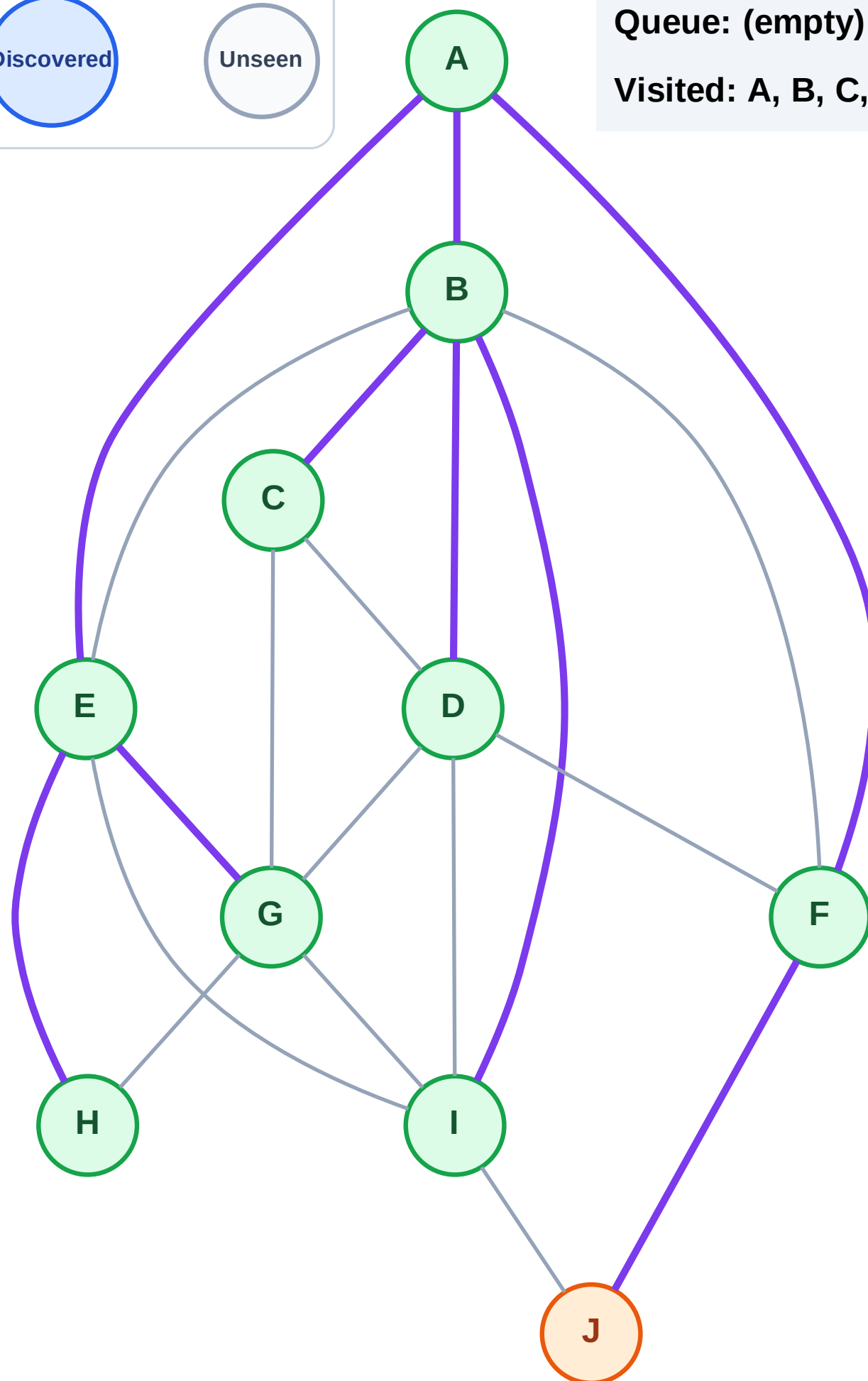
Step 20: Process node J

## Legend



Queue: (empty)

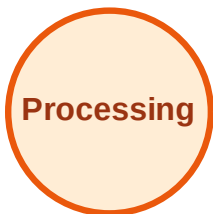
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

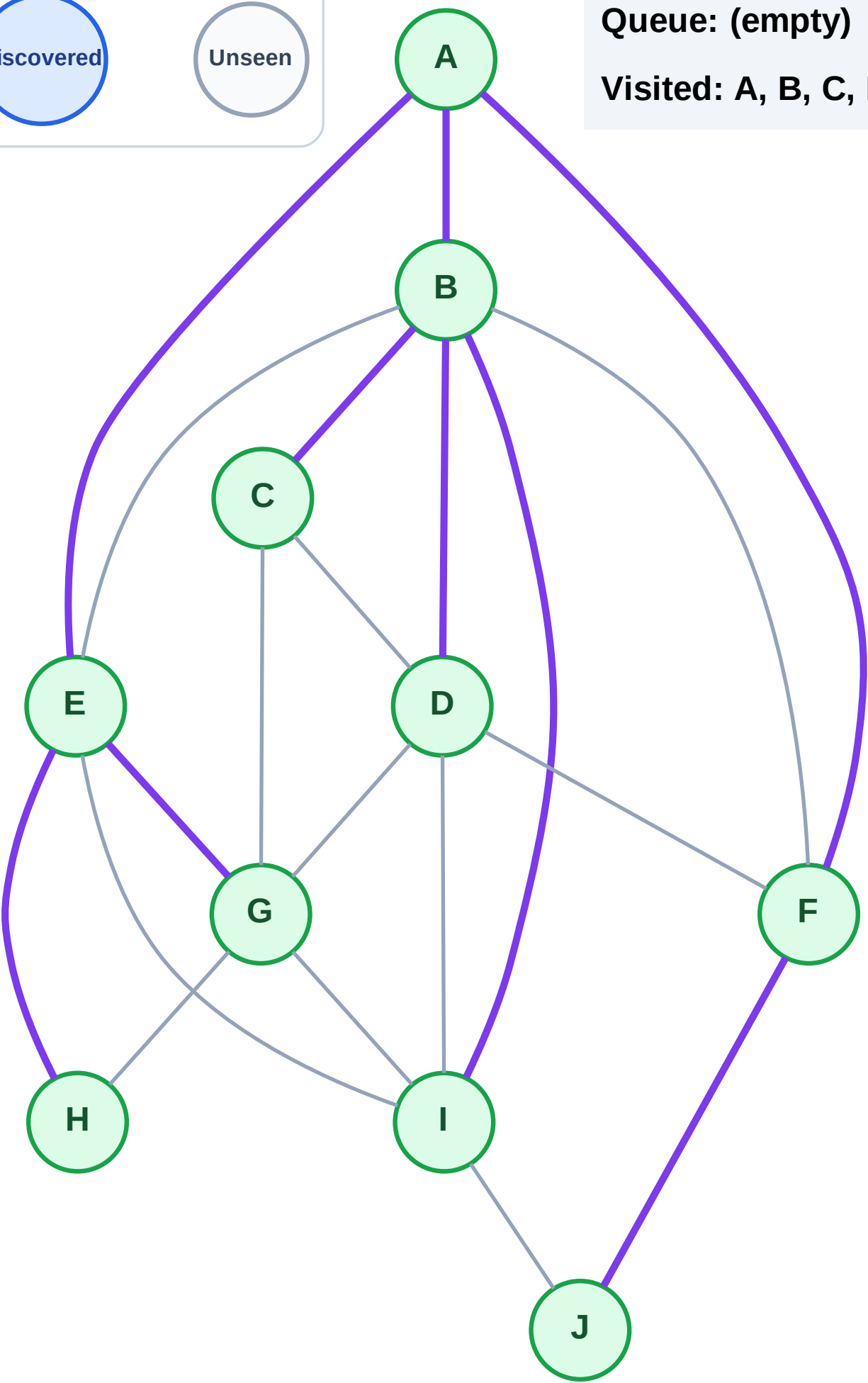
Step 21: No new neighbors from J

## Legend



Queue: (empty)

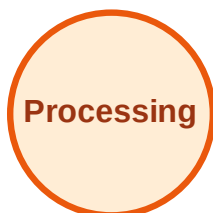
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

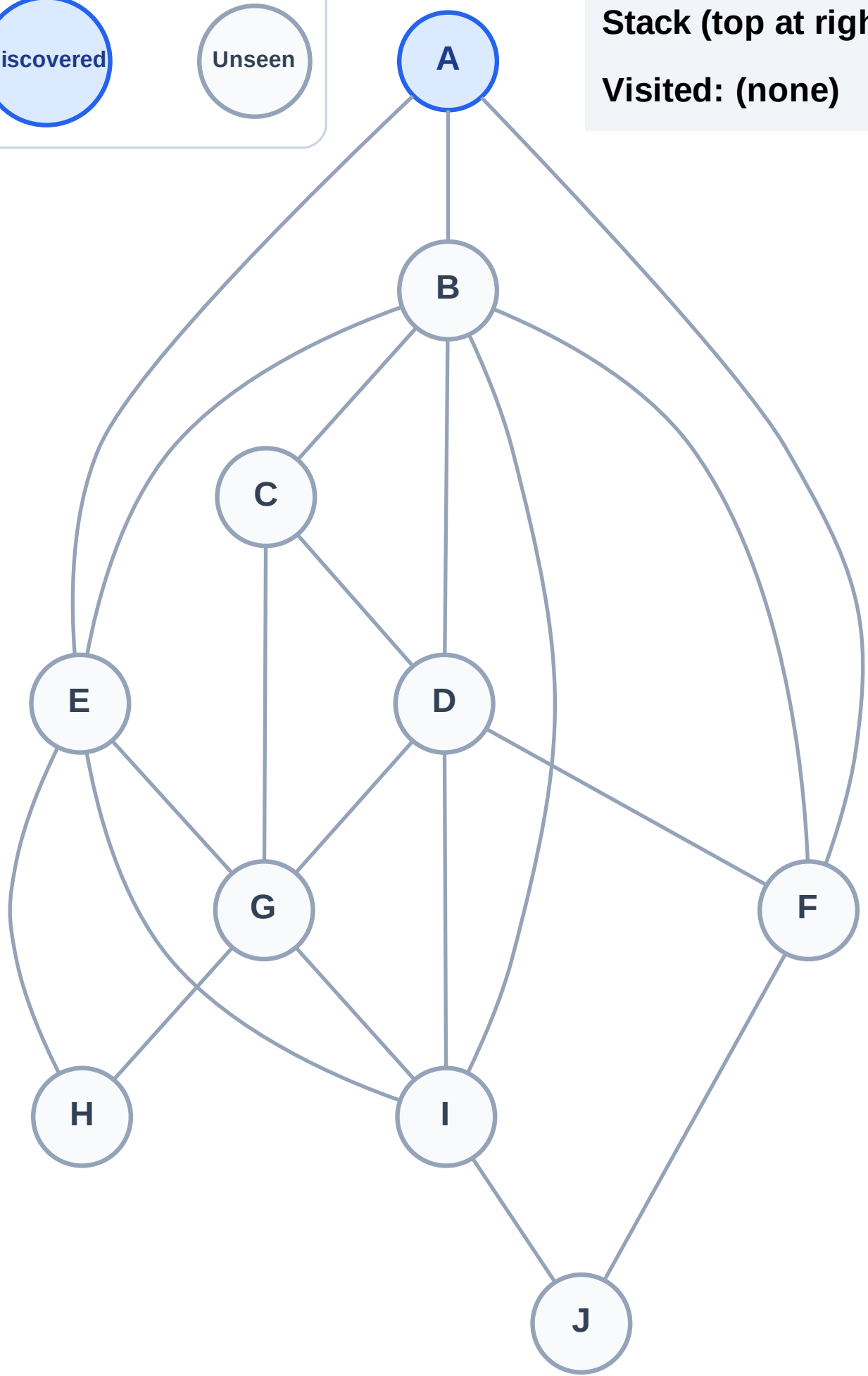
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

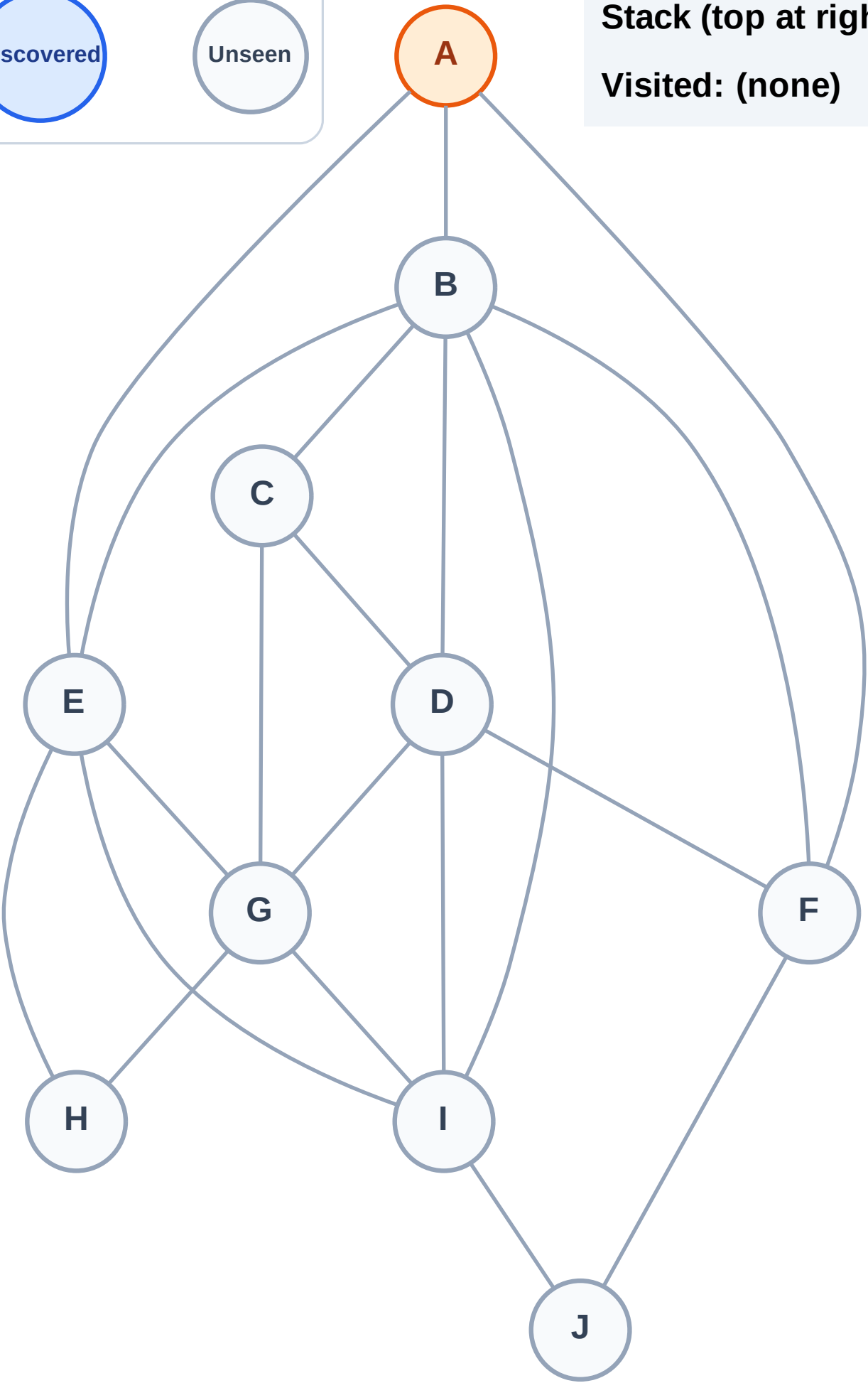
Complete

Processing

Discovered

Unseen

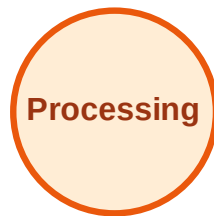
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

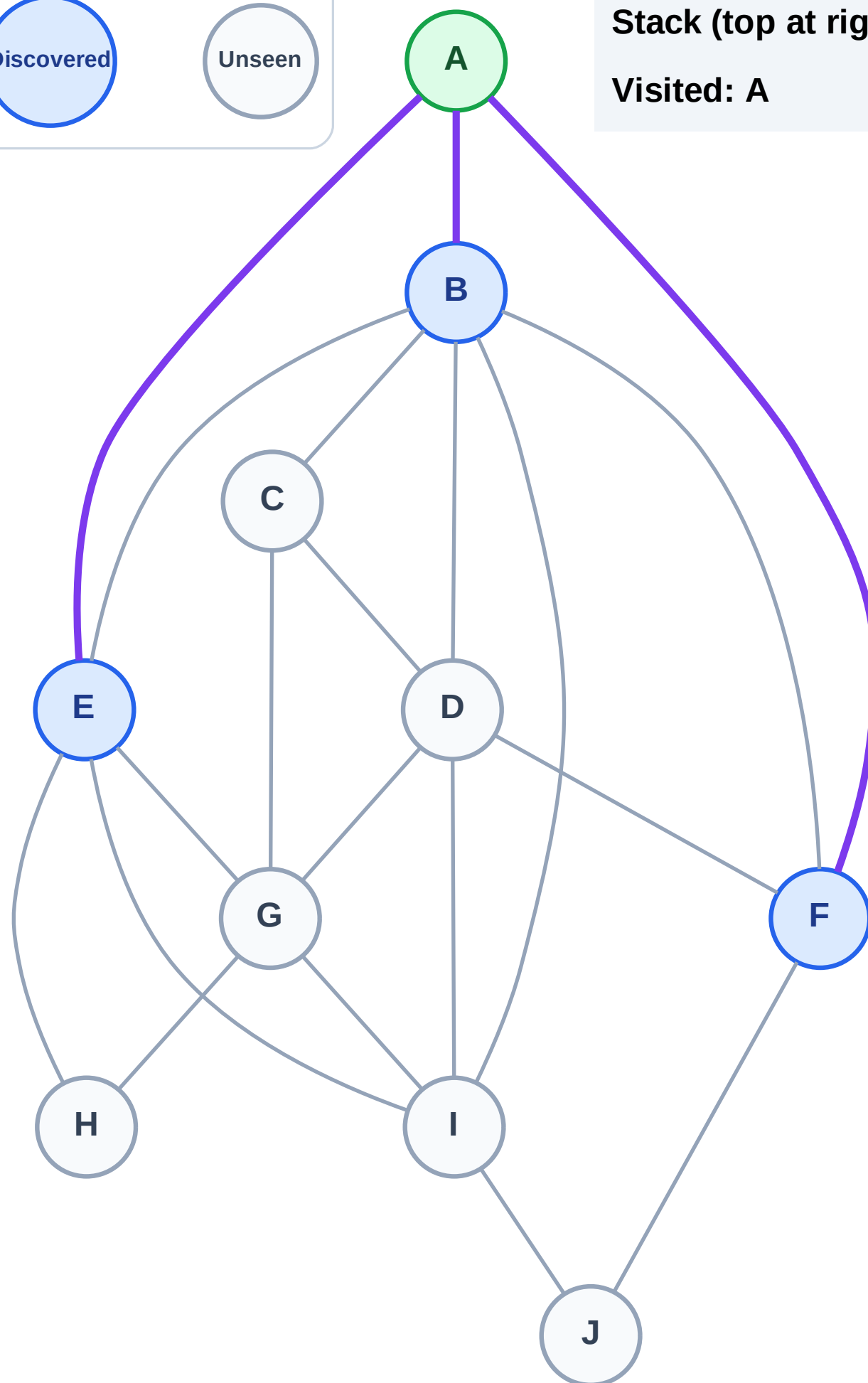
Step 3: Discover F, E, B from A

## Legend



Stack (top at right): F | E | B

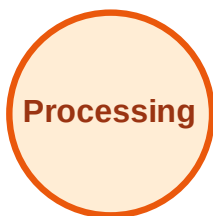
Visited: A



# DFS Traversal

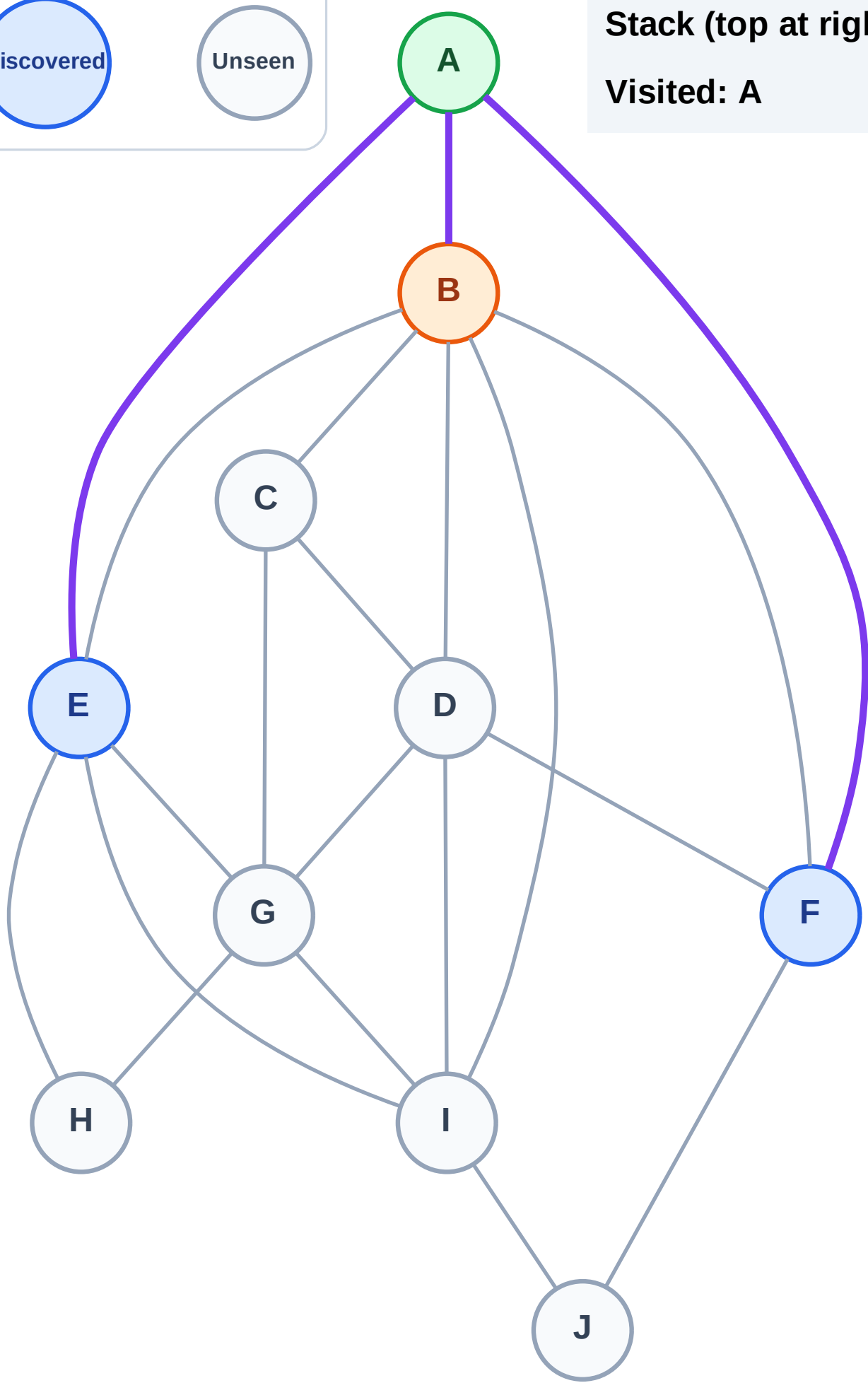
Step 4: Process node B

## Legend



Stack (top at right): F | E

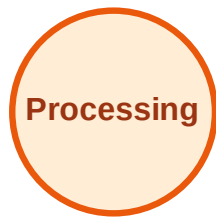
Visited: A



# DFS Traversal

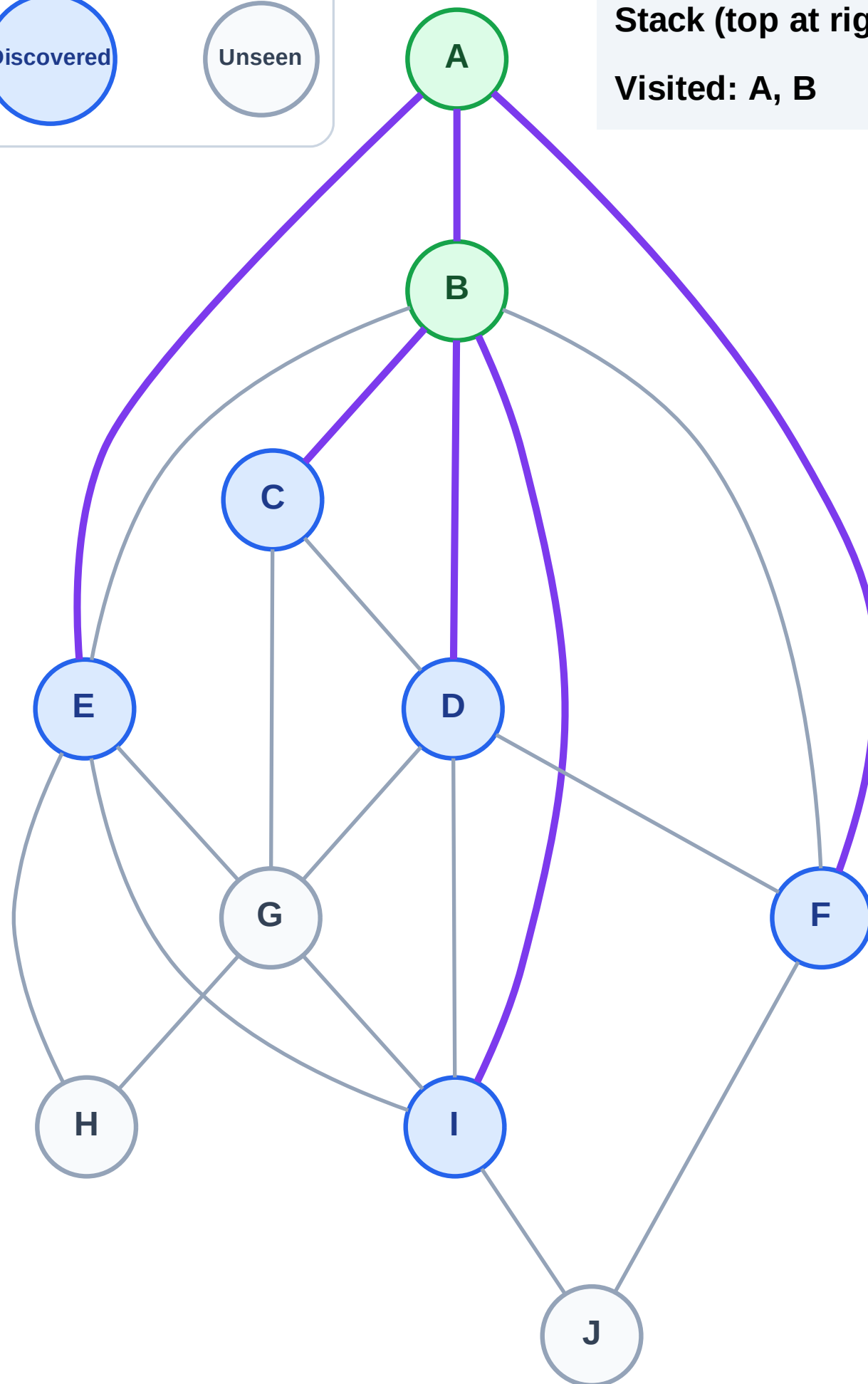
Step 5: Discover I, D, C from B

## Legend



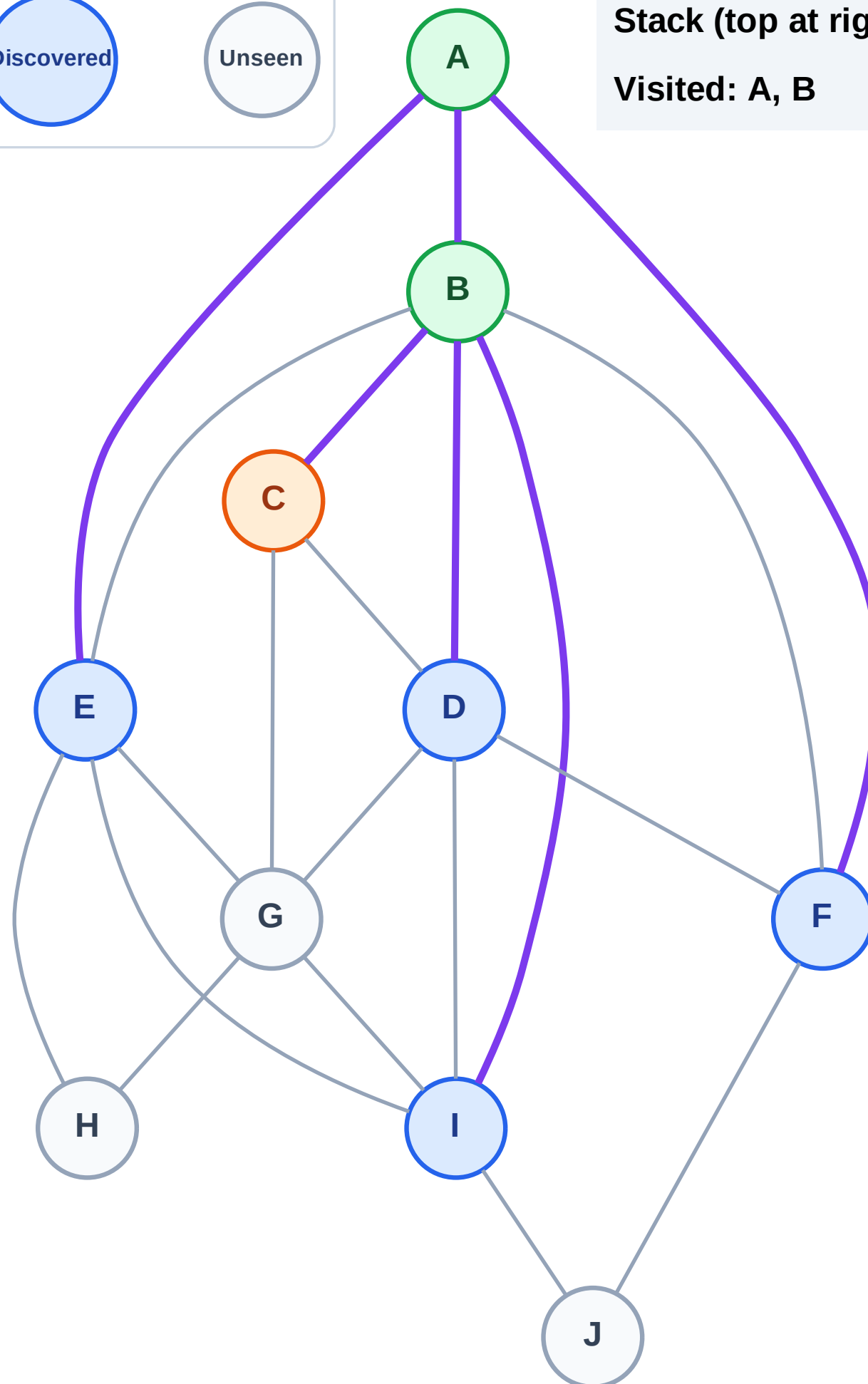
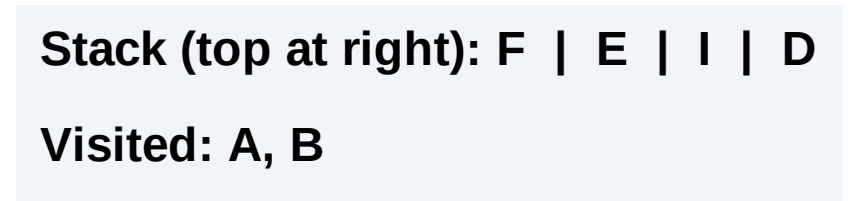
Stack (top at right): F | E | I | D | C

Visited: A, B



### Step 6: Process node C

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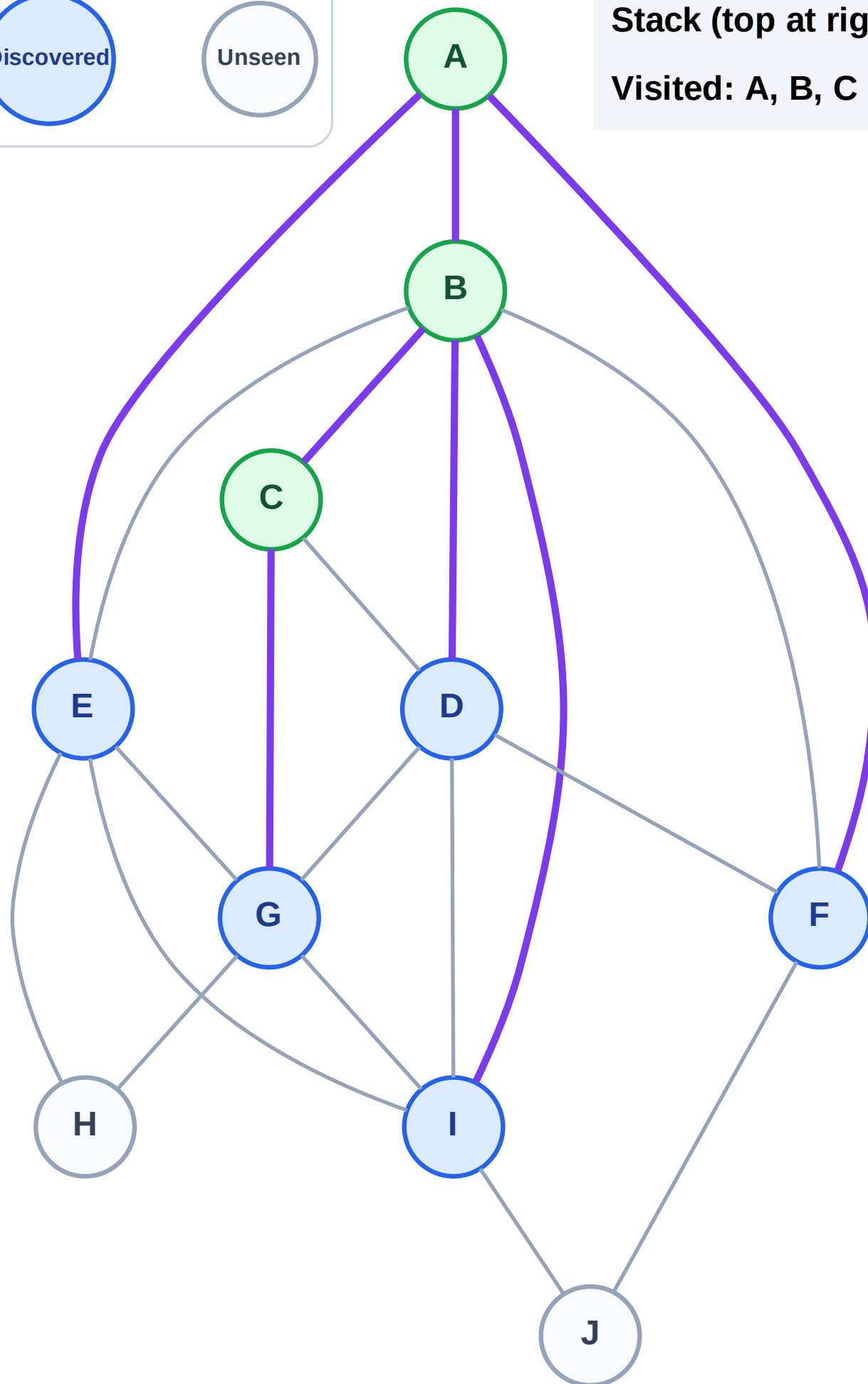


### Step 7: Discover G from C

### Legend

## Unseen

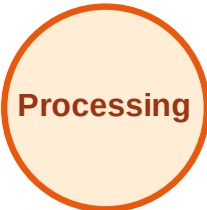
**Visited: A, B, C**



# DFS Traversal

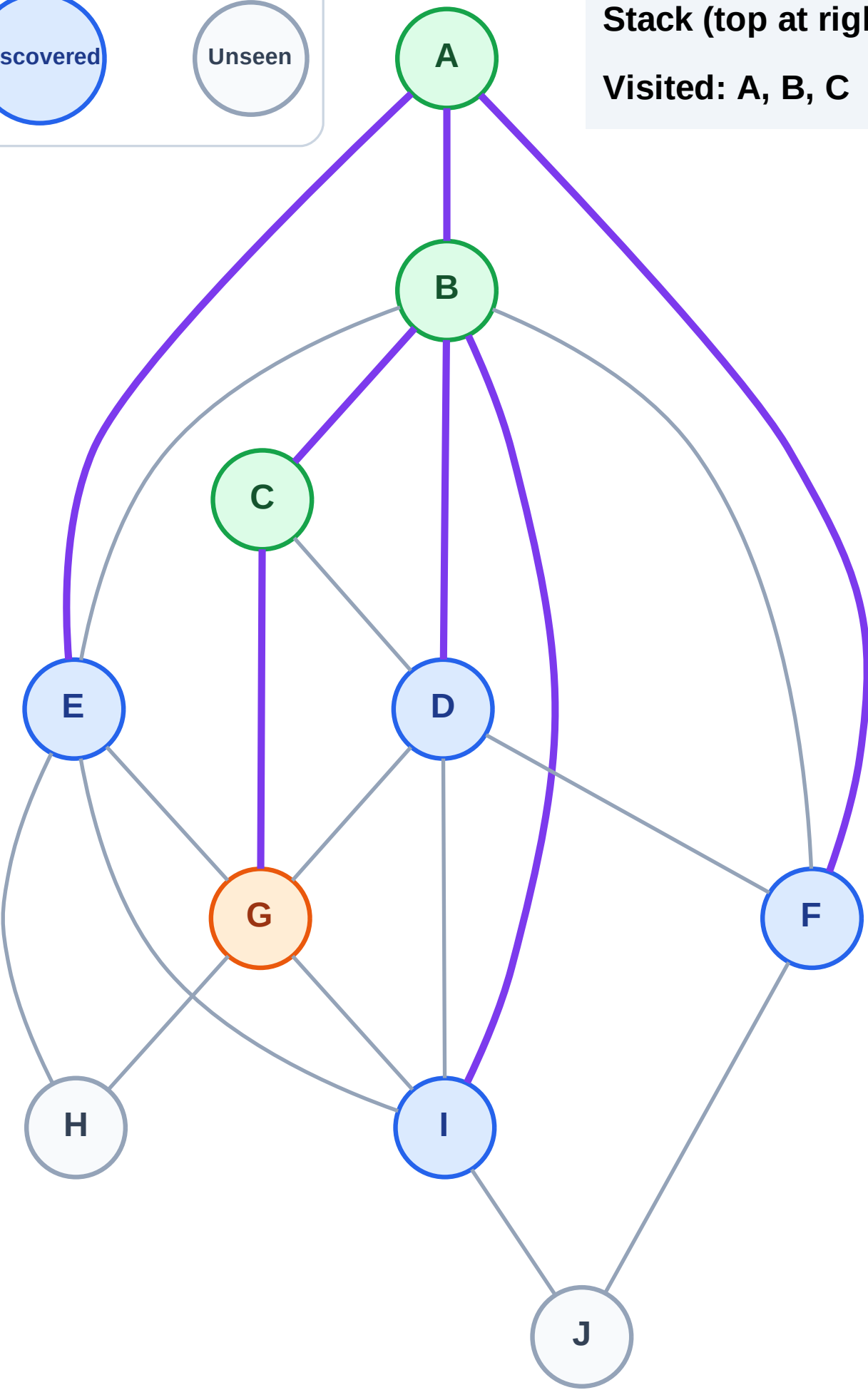
Step 8: Process node G

## Legend



Stack (top at right): F | E | I | D

Visited: A, B, C



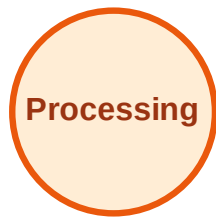
# DFS Traversal

Step 9: Discover H from G

## Legend



Complete



Processing



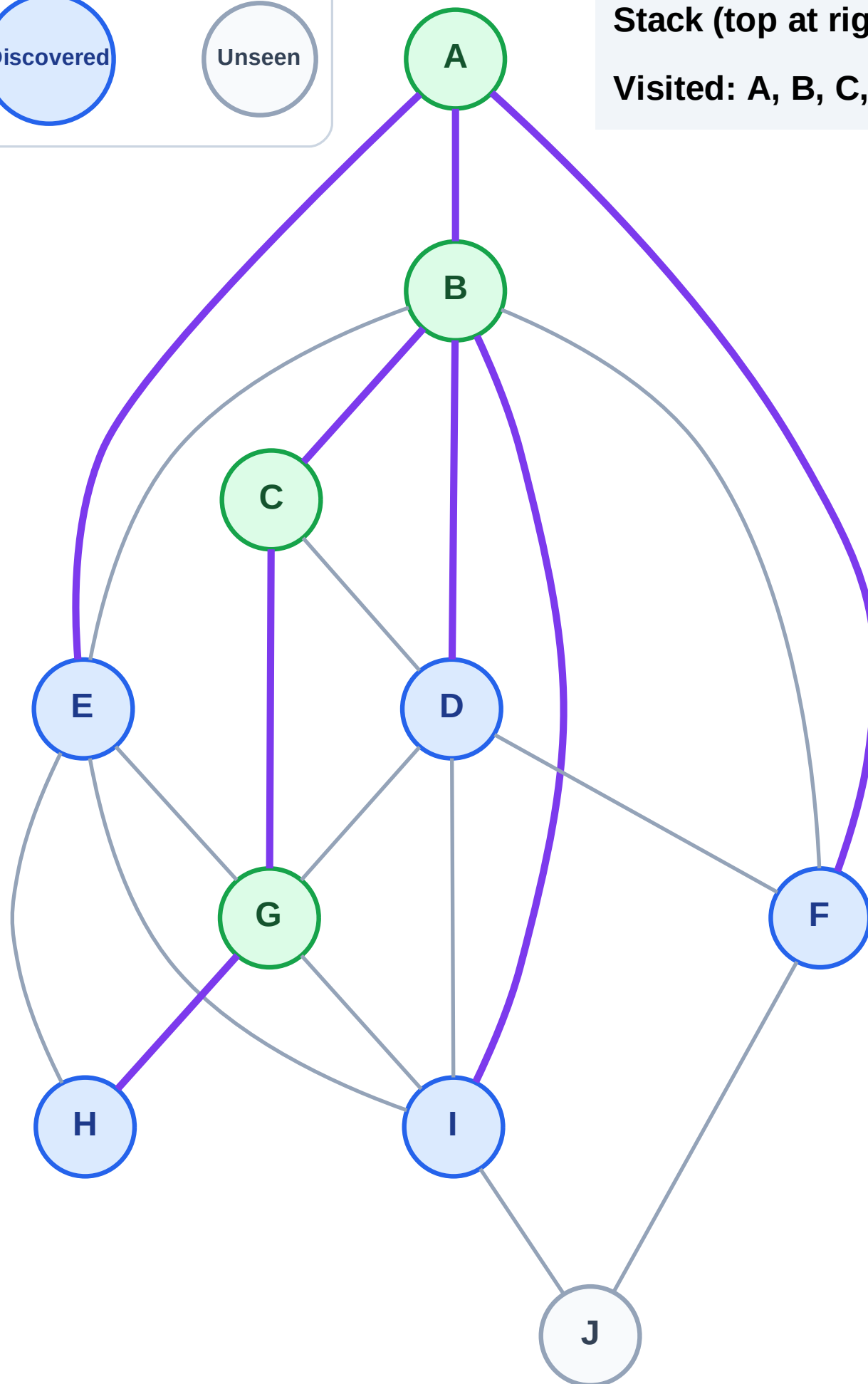
Discovered



Unseen

Stack (top at right): F | E | I | D | H

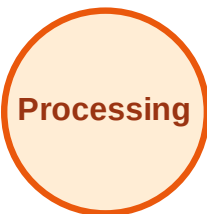
Visited: A, B, C, G



# DFS Traversal

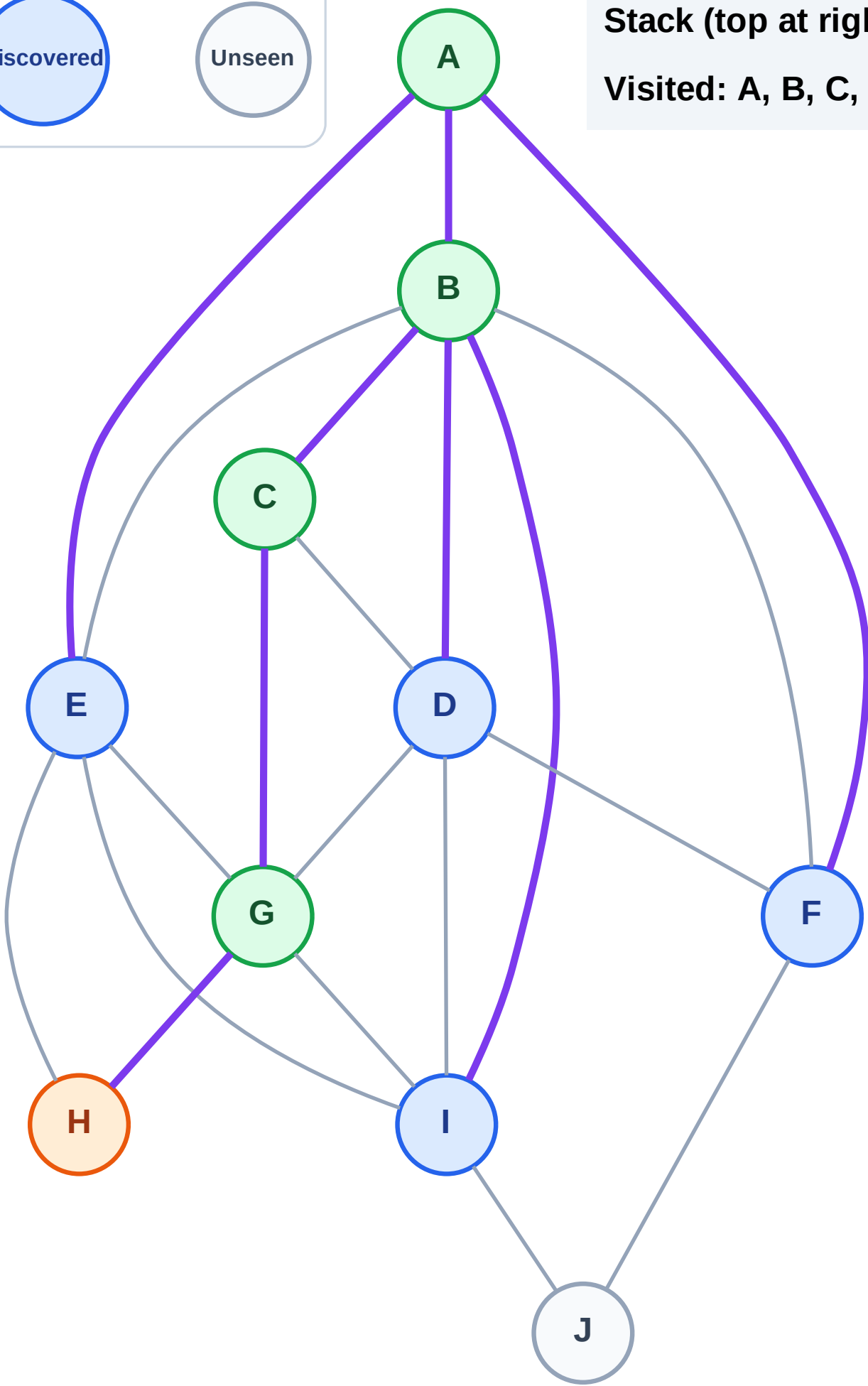
Step 10: Process node H

## Legend



Stack (top at right): F | E | I | D

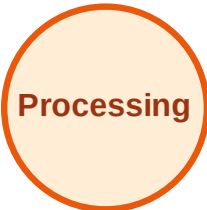
Visited: A, B, C, G



# DFS Traversal

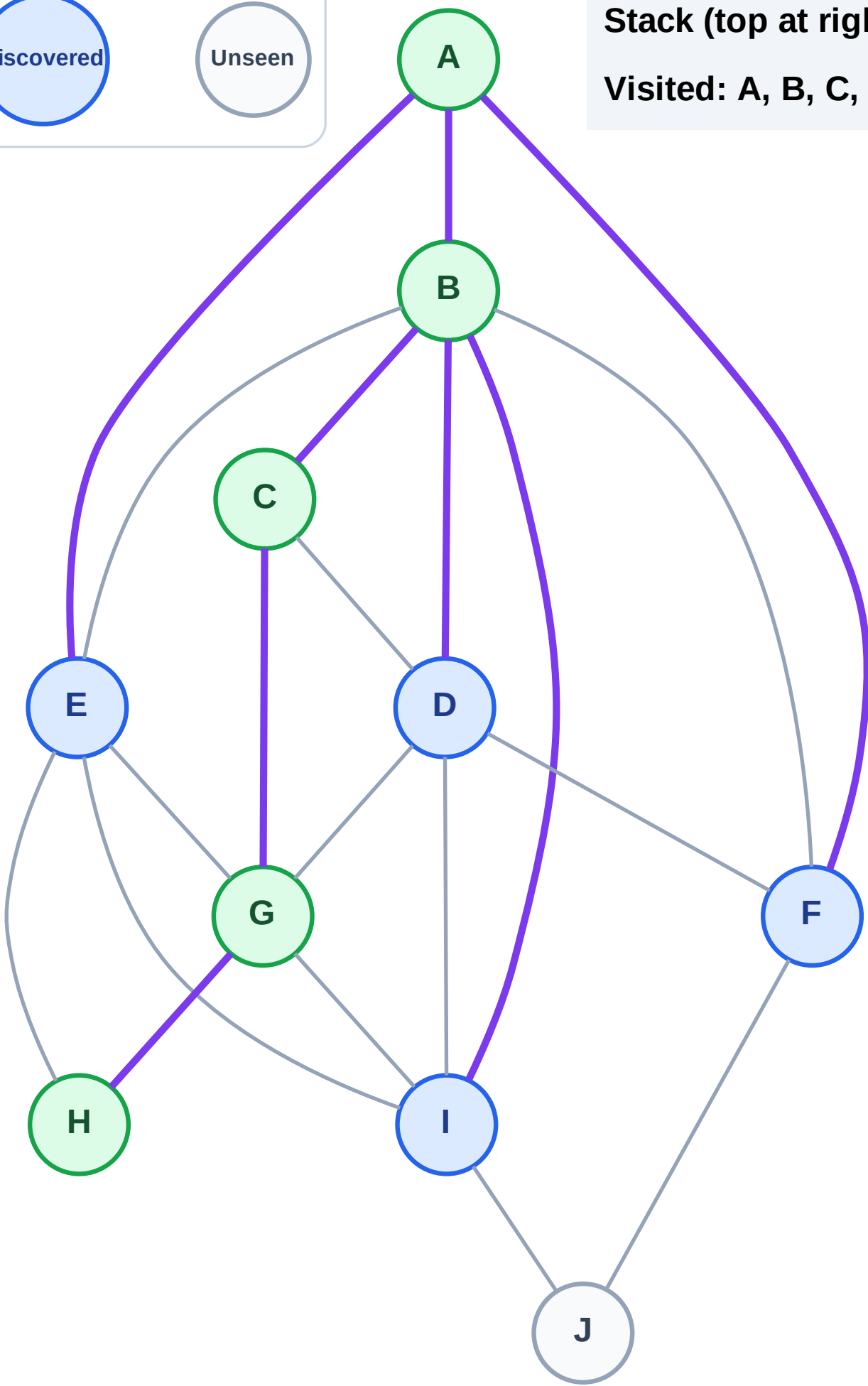
Step 11: No new neighbors from H

## Legend



Stack (top at right): F | E | I | D

Visited: A, B, C, G, H



# DFS Traversal

Step 12: Process node D

## Legend

Complete

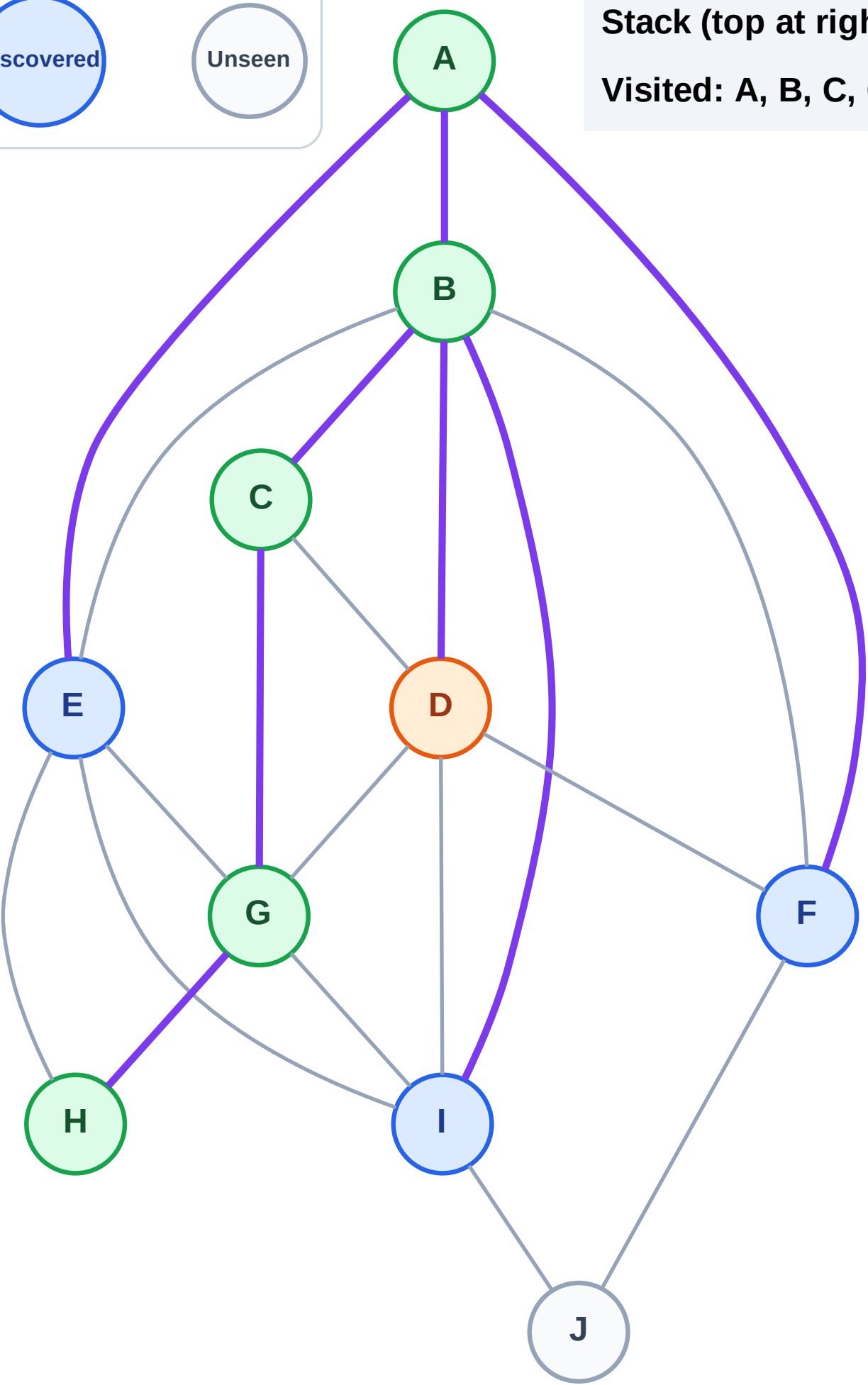
Processing

Discovered

Unseen

Stack (top at right): F | E | I

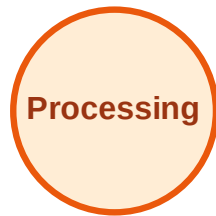
Visited: A, B, C, G, H



# DFS Traversal

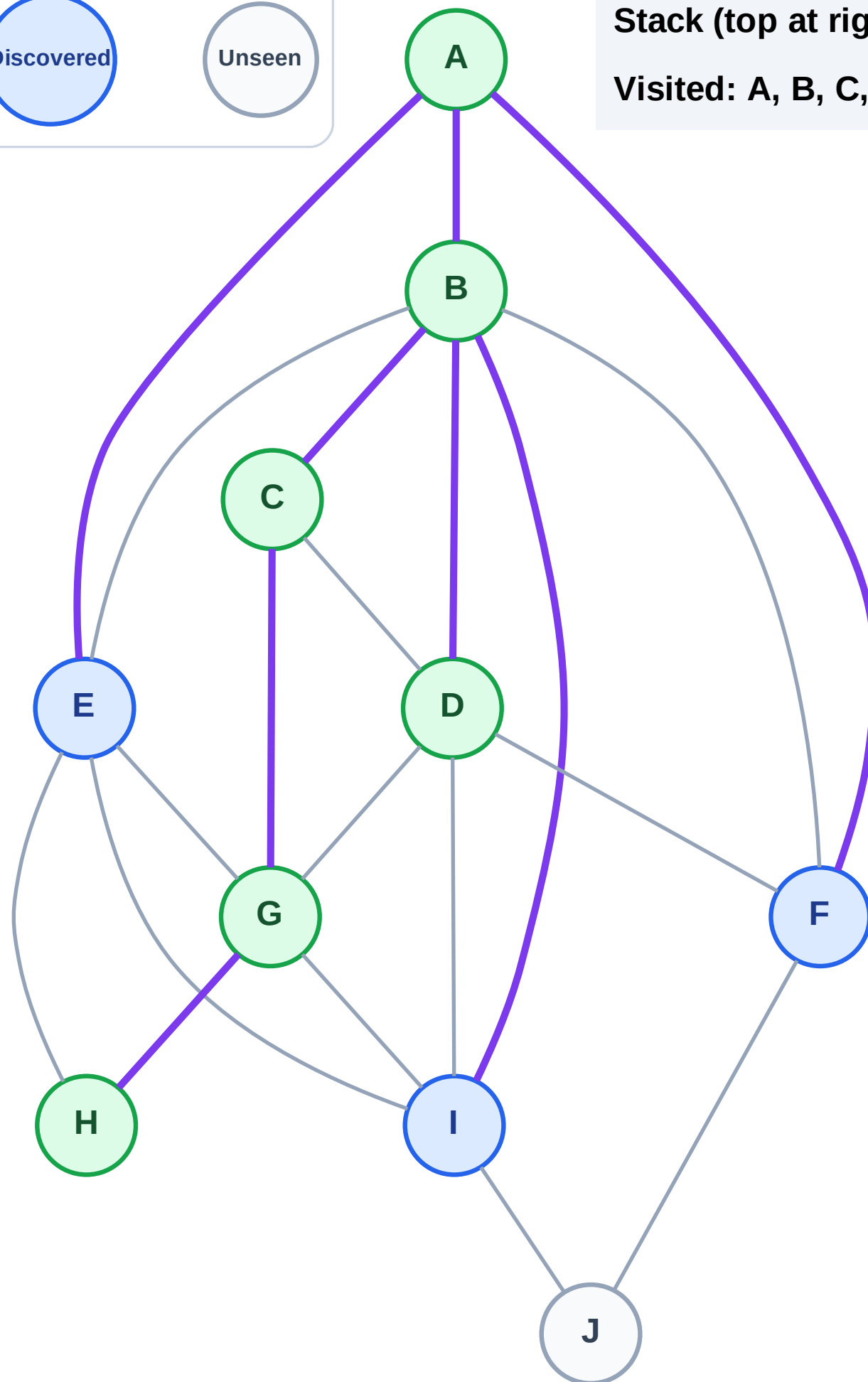
Step 13: No new neighbors from D

## Legend



Stack (top at right): F | E | I

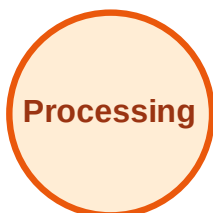
Visited: A, B, C, D, G, H



# DFS Traversal

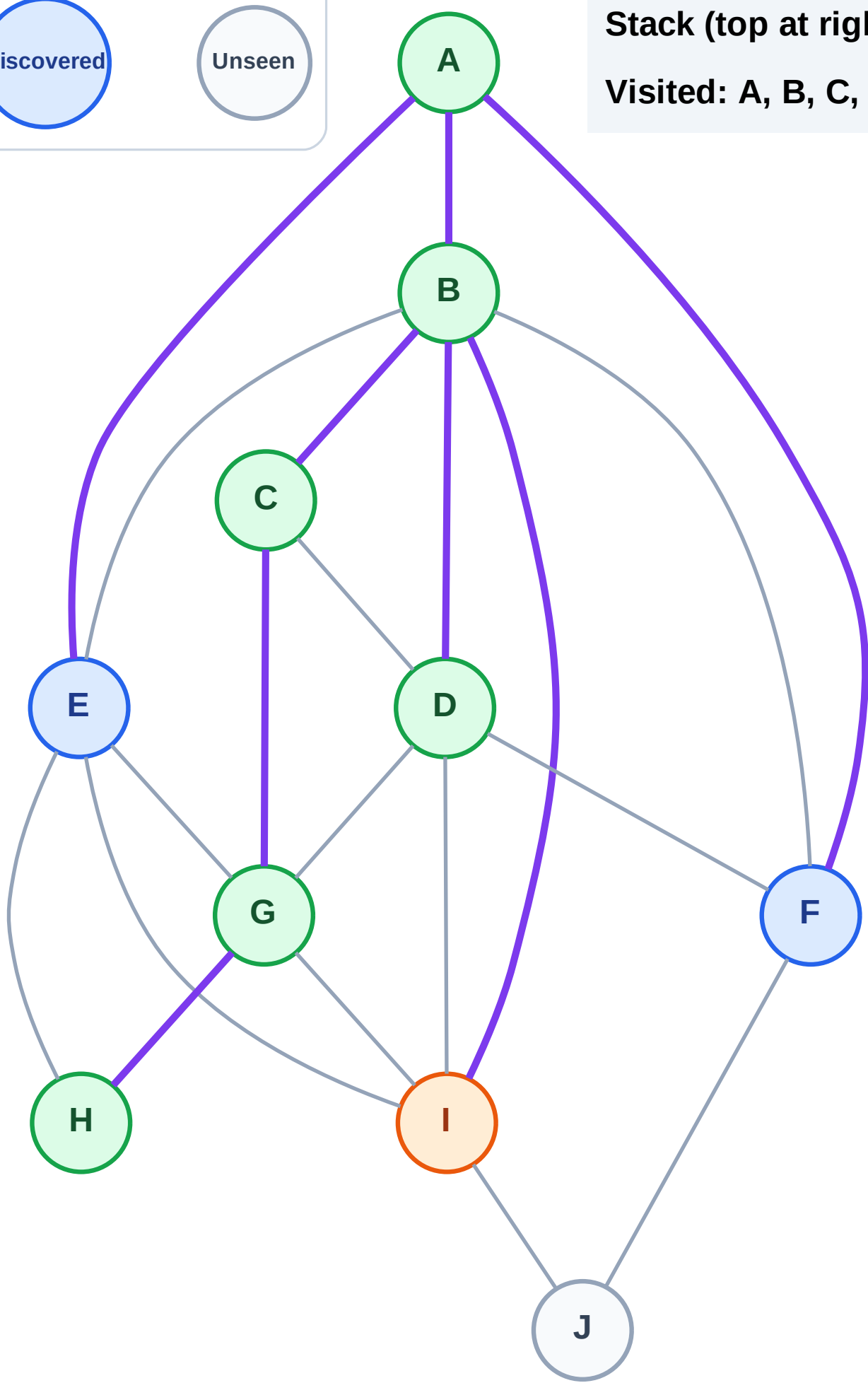
Step 14: Process node I

## Legend



Stack (top at right): F | E

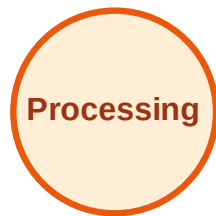
Visited: A, B, C, D, G, H



# DFS Traversal

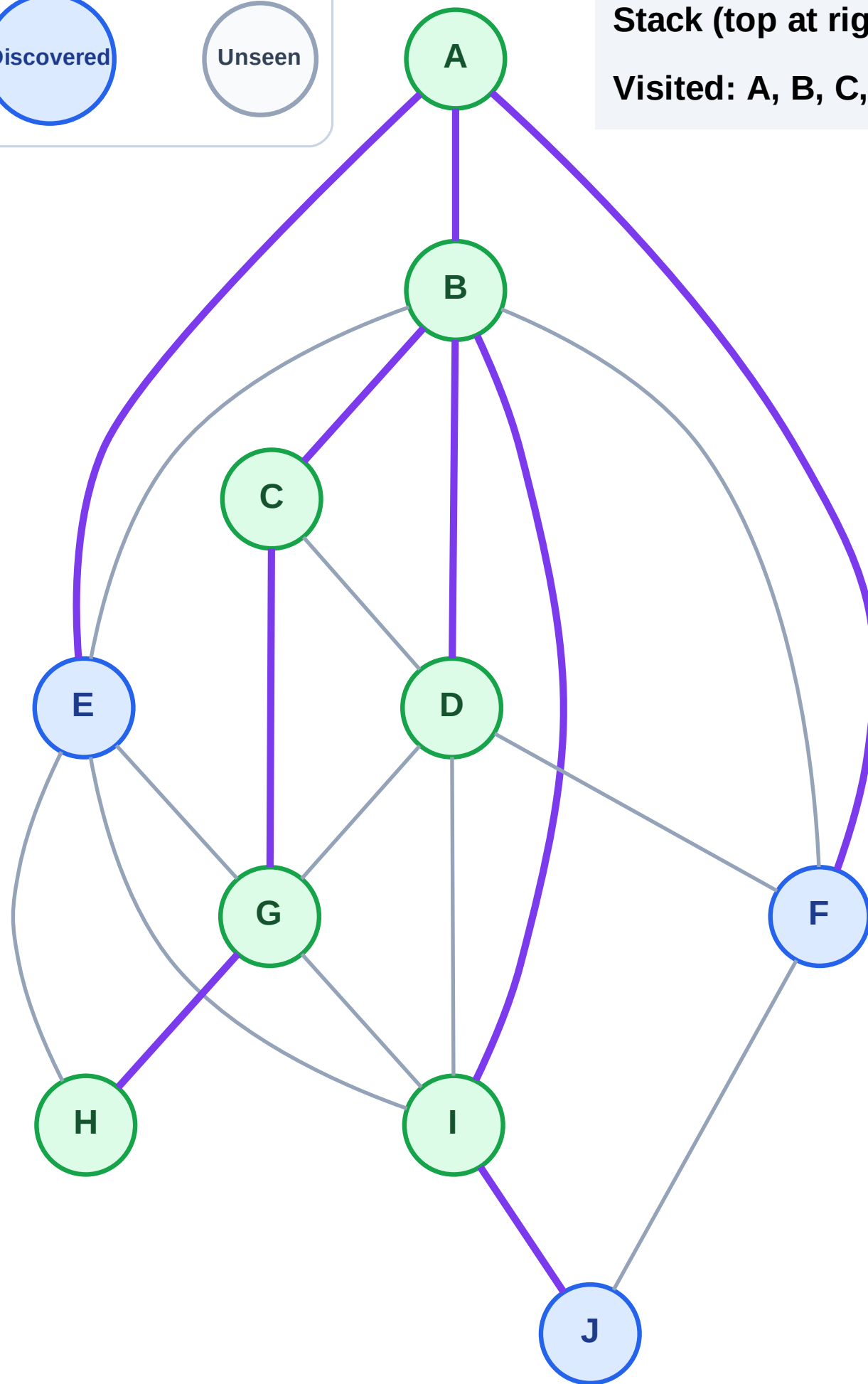
Step 15: Discover J from I

## Legend



Stack (top at right): F | E | J

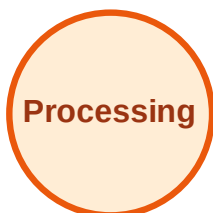
Visited: A, B, C, D, G, H, I



# DFS Traversal

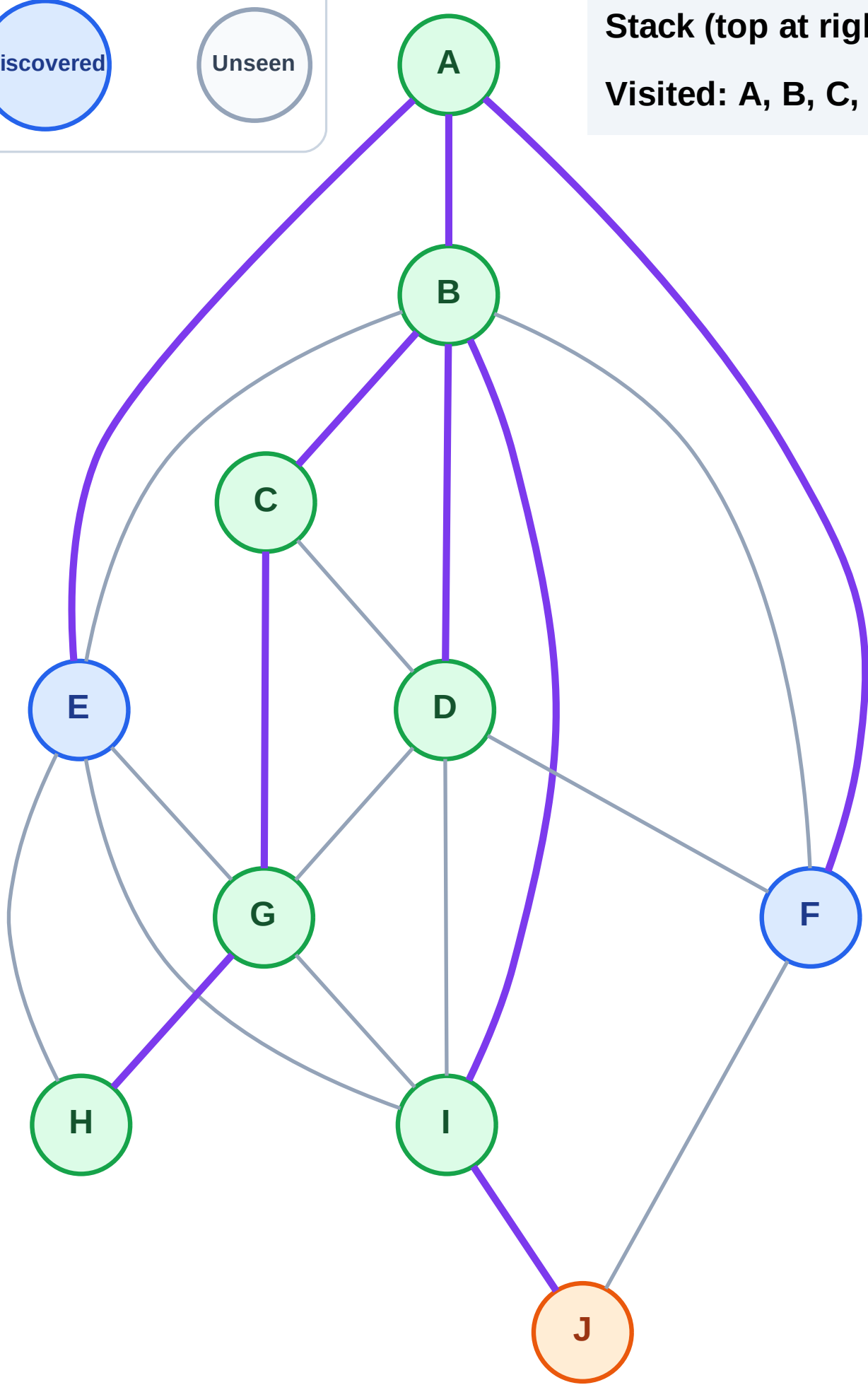
Step 16: Process node J

## Legend



Stack (top at right): F | E

Visited: A, B, C, D, G, H, I



# DFS Traversal

Step 17: No new neighbors from J

## Legend

Complete

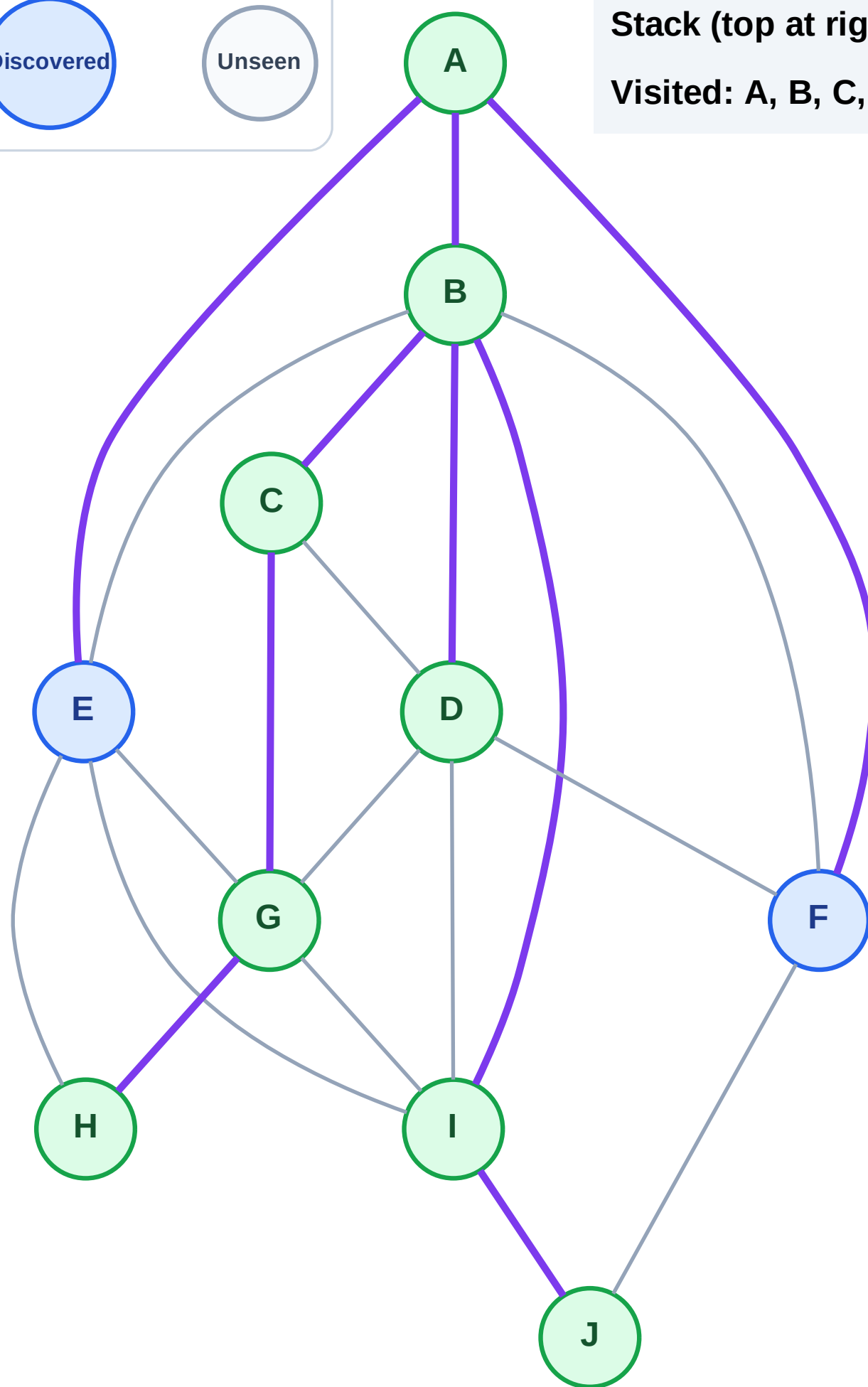
Processing

Discovered

Unseen

Stack (top at right): F | E

Visited: A, B, C, D, G, H, I, J



# DFS Traversal

Step 18: Process node E

Legend

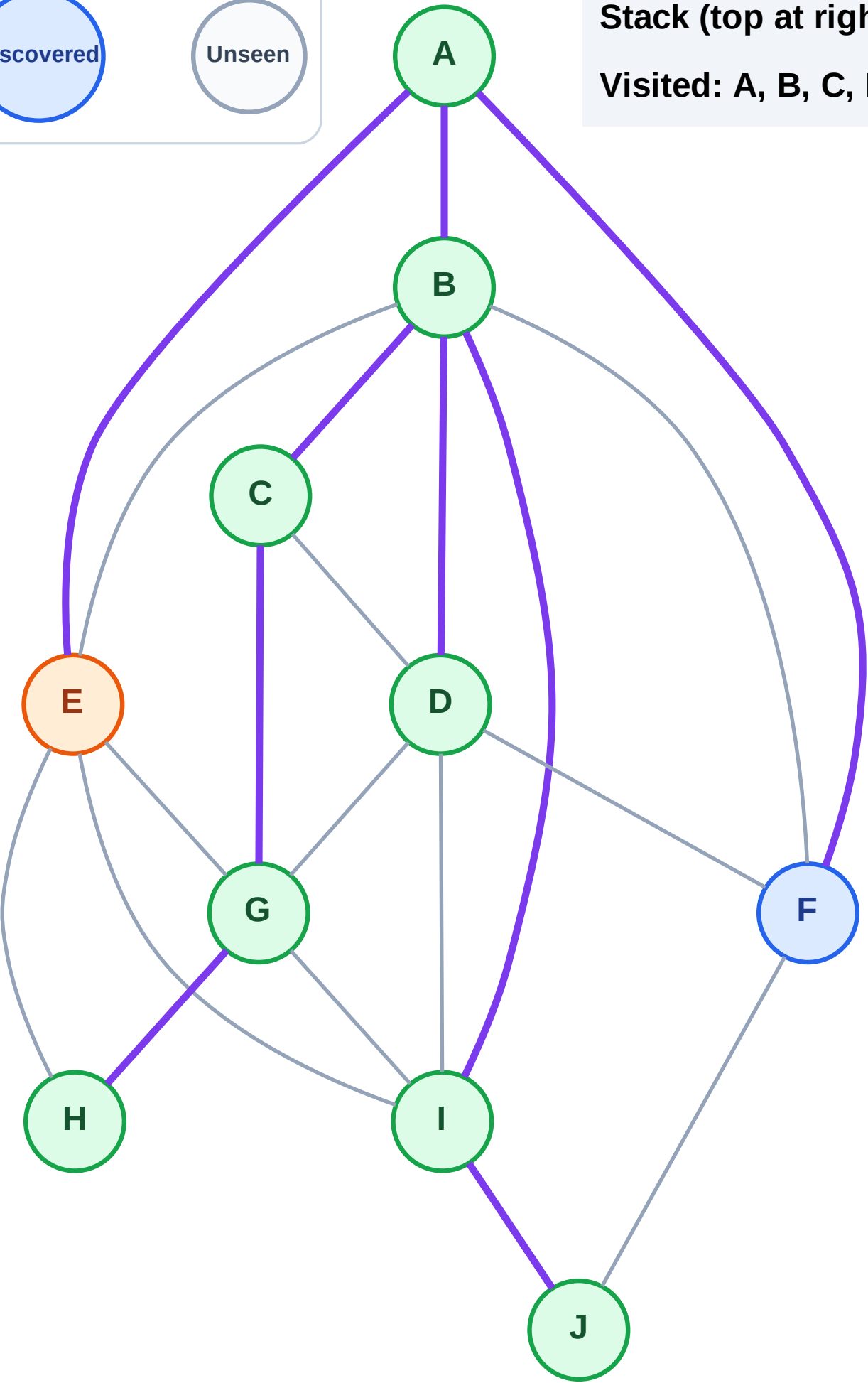
Complete

Processing

Discovered

Unseen

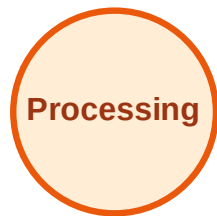
Stack (top at right): F  
Visited: A, B, C, D, G, H, I, J



# DFS Traversal

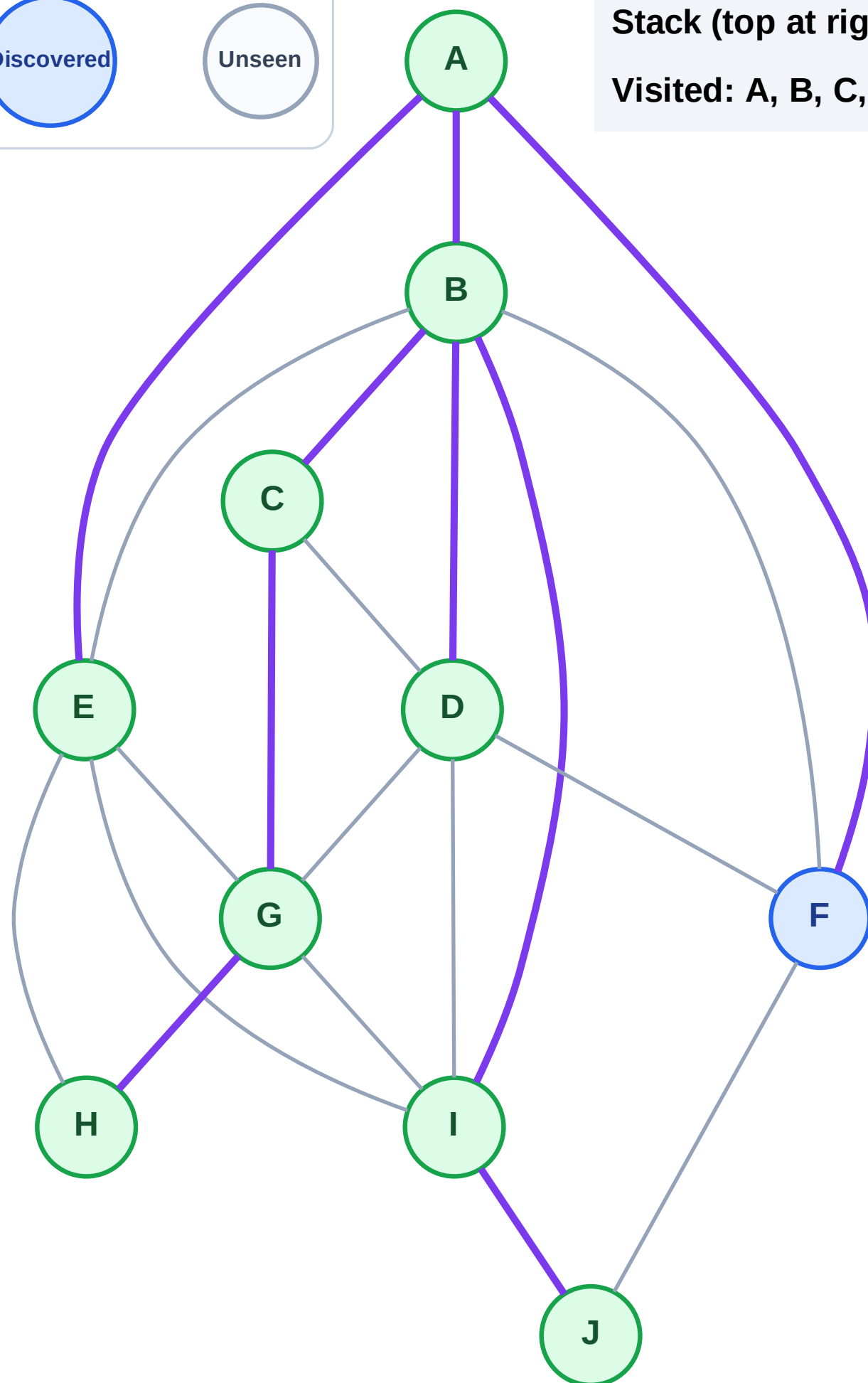
Step 19: No new neighbors from E

## Legend



Stack (top at right): F

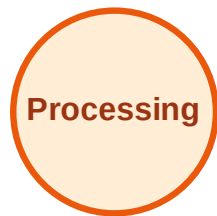
Visited: A, B, C, D, E, G, H, I, J



# DFS Traversal

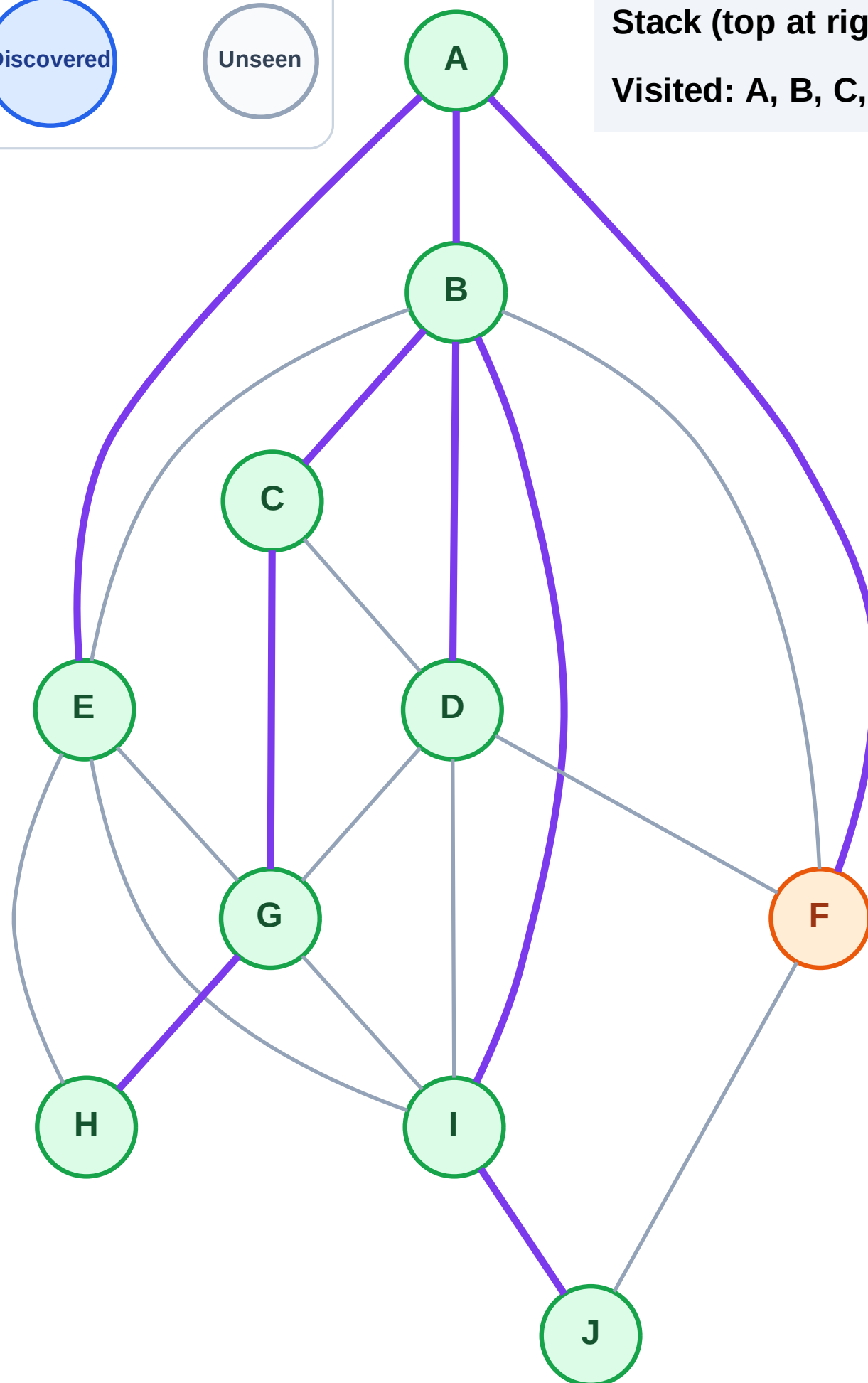
Step 20: Process node F

## Legend



Stack (top at right): (empty)

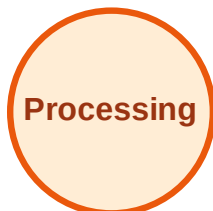
Visited: A, B, C, D, E, G, H, I, J



# DFS Traversal

Step 21: No new neighbors from F

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

